EXPENSEIQ

AI-Powered Business Expense Management System

PROJECT DOCUMENTATION, IMPLEMENTATION & TESTING REPORT

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Course / Programme: AI Automation Diploma

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Document Control

Item	Details
Project	ExpenseIQ — AI-Powered Business Expense Management System
Prepared by	Toluwanimi Akinyemi
Programme	AI Automation Diploma
Instructor	Ogbu Princewill
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Business Context	Nigerian small-business expense management
Currency	NGN
Primary Database	Supabase
Automation Platform	Make.com
Frontend	Lovable
Version	1.0 (final submission)
AI Model	NVIDIA Nemotron 3 Ultra (free) via OpenRouter
Reporting	Airtable mirror and Google Sheets weekly summary
Notifications	Gmail
Live Application	https://expense-ai-pro-14.lovable.app/login

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1. Executive Summary

ExpenseIQ is an AI-assisted business expense management system designed for a Nigerian small-business environment. It provides a web interface for employees to submit expenses and receipts, a role-based interface for managers and administrators to review expenses, and an automation layer for categorisation, approval routing, notifications and weekly reporting.

The system combines Lovable for the frontend, Supabase as the primary database and source of truth, and Make.com for background automation. OpenRouter provides AI categorisation, Google Drive provides durable receipt storage, Airtable provides a downstream reporting mirror, Gmail provides notifications, and Google Sheets provides weekly reporting.

AI is restricted to suggestions: category, subcategory, summary, duplicate risk and reviewer note. It does not approve or reject expenses, make payment decisions, make tax or legal conclusions, or override the approval threshold. Human managers remain responsible for financial decisions.

The solution has been implemented and tested end to end. Expenses submitted through Lovable are stored in Supabase, categorised by AI, routed by the NGN 100,000 approval rule, decided by a manager, notified to the employee by email, mirrored to Airtable and summarised each week in Google Sheets and a management email. Screenshots of every stage are included in this report and are listed in Section 14. The live application is available at the link in Document Control and Appendix A.6.

2. Project Objectives

- Provide employees with a simple interface for submitting business expenses.
- Capture supporting receipts while allowing submission to continue when a receipt is missing.
- Store expense records centrally in Supabase.
- Use AI to suggest fixed expense categories and supporting review information.
- Automatically identify expenses requiring manager approval.
- Notify managers when approval is required.
- Allow managers to approve, reject, or request clarification.
- Notify employees of manager decisions.
- Preserve original submitted information and append decision information.
- Mirror processed expenses to Airtable for management reporting.
- Generate weekly expense summaries by status, category and department.
- Reduce manual and repetitive expense-processing activities.

3. Business Problem and Proposed Solution

3.1 Business Problem

Expense processing can involve repetitive data entry, receipt handling, manual categorisation, approval routing, notifications and reporting. ExpenseIQ addresses these repetitive activities through a connected web application and automated workflow.

3.2 Proposed Solution

Employees submit expenses through Lovable. Supabase stores the authoritative record and triggers Make.com. Make.com processes the receipt, obtains AI suggestions, updates Supabase, mirrors the record to

Airtable, and applies the explicit approval rule. Manager decisions are written to Supabase and trigger a second Make.com workflow for employee notification.

4. System Architecture

Layer	Component	Primary Responsibility
Frontend	Lovable	Employee submission and manager/admin dashboards
Database	Supabase	Authoritative expense and user data; webhook source
Automation	Make.com	Workflow orchestration, routing and notifications
AI	OpenRouter	Expense categorisation suggestions
File Storage	Google Drive	Durable receipt storage
Reporting Mirror	Airtable	Downstream management/backup mirror
Notifications	Gmail	Approval and status emails
Reporting	Google Sheets	Weekly summary reporting

Architecture flow:

Employee → Lovable → Supabase → Database Webhook → Make.com → AI / Drive / Airtable / Gmail / Google Sheets

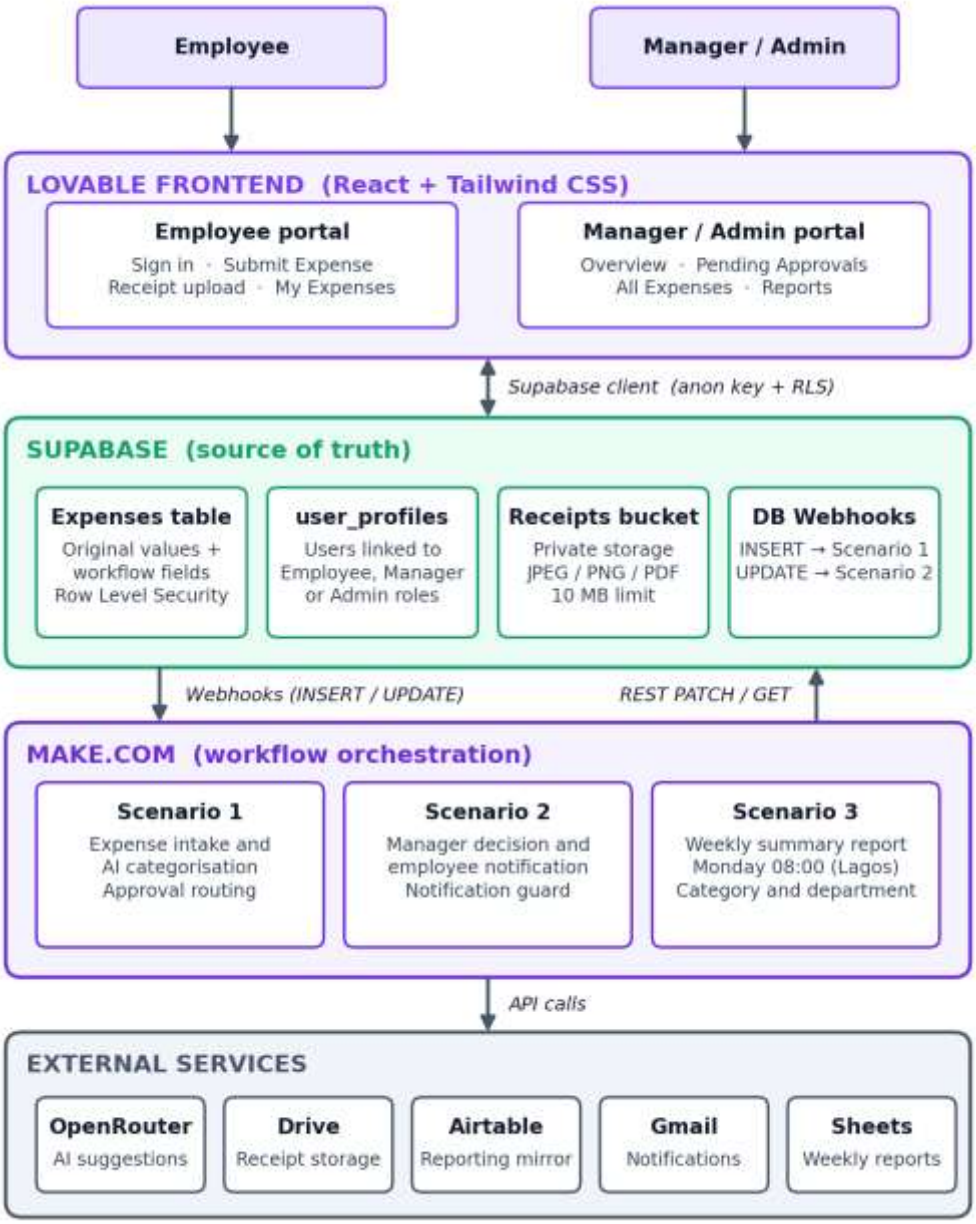


Figure 1: ExpenseIQ system architecture showing the frontend, database, automation and external-service layers

Figure 1 shows how a request moves through the system:

- **Submit.** The employee signs in to the Lovable frontend and submits an expense with an optional receipt. The record is saved in Supabase and the receipt in the private receipts bucket.
- **Trigger.** A Supabase Database Webhook fires on every INSERT and sends the new record to Scenario 1 in Make.com.
- **Process.** Scenario 1 copies the receipt to Google Drive, obtains AI suggestions from OpenRouter, writes the results back to Supabase and mirrors the record to Airtable.
- **Route.** The router applies the NGN 100,000 rule. Expenses at or above the threshold go to the manager by Gmail; smaller expenses are recorded and the employee is sent a confirmation.

- **Decide.** The manager approves, rejects or requests clarification in the dashboard. The status change fires the UPDATE webhook and Scenario 2 notifies the employee.
- **Report.** Scenario 3 reads the last seven days of expenses from Supabase, writes the weekly, category and department summaries to Google Sheets and emails management.

5. Technology Stack

Technology	Use
Lovable	Frontend web application
React + Tailwind CSS	Frontend implementation
Supabase	Database, authentication, storage integration and webhooks
Make.com	Automation and orchestration
OpenRouter	AI categorisation
Google Drive	Durable receipt storage
Airtable	Downstream reporting mirror
Gmail	Notifications
Google Sheets	Weekly reporting

6. Supabase Database Implementation

6.1 Expenses Table

The expenses table stores original submission information and workflow-generated fields.

Field Group	Examples / Purpose
Identity	id, expense_ref
Employee	employee_name, employee_email, department
Expense	expense_date, vendor, amount, purpose, payment_method, project_client, additional_notes
Receipt	receipt_url, receipt_missing
AI	ai_category, ai_subcategory, ai_summary, duplicate_risk, ai_reviewer_note
Workflow	approval_required, status, manager_notes, notification_sent
Timestamps	submitted_at, approved_at, created_at, updated_at

6.2 User Profiles and Roles

The user_profiles table links authenticated users to employee, manager or admin roles.

6.3 Row Level Security

RLS is enabled on expenses. Employees are restricted to their own records, while managers and administrators can access records according to the defined policies.

6.4 Receipt Storage

The receipts bucket is private, supports JPEG/PNG/PDF files, and has a 10 MB limit.

6.5 Database Webhooks

An INSERT webhook triggers Scenario 1. An UPDATE/status-change webhook triggers Scenario 2. Database Webhooks are used instead of Realtime.

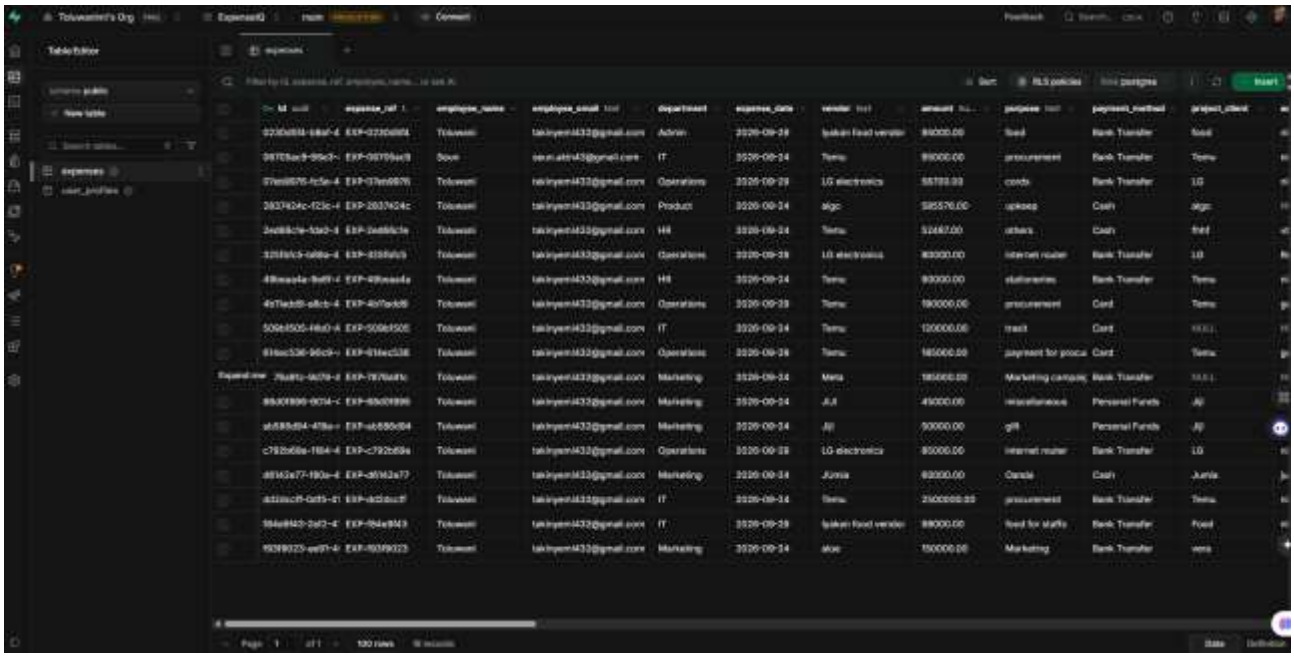


Figure 2: Supabase expenses table showing submitted expense records and their generated references



Figure 3: Supabase user_profiles table showing Employee and Manager roles

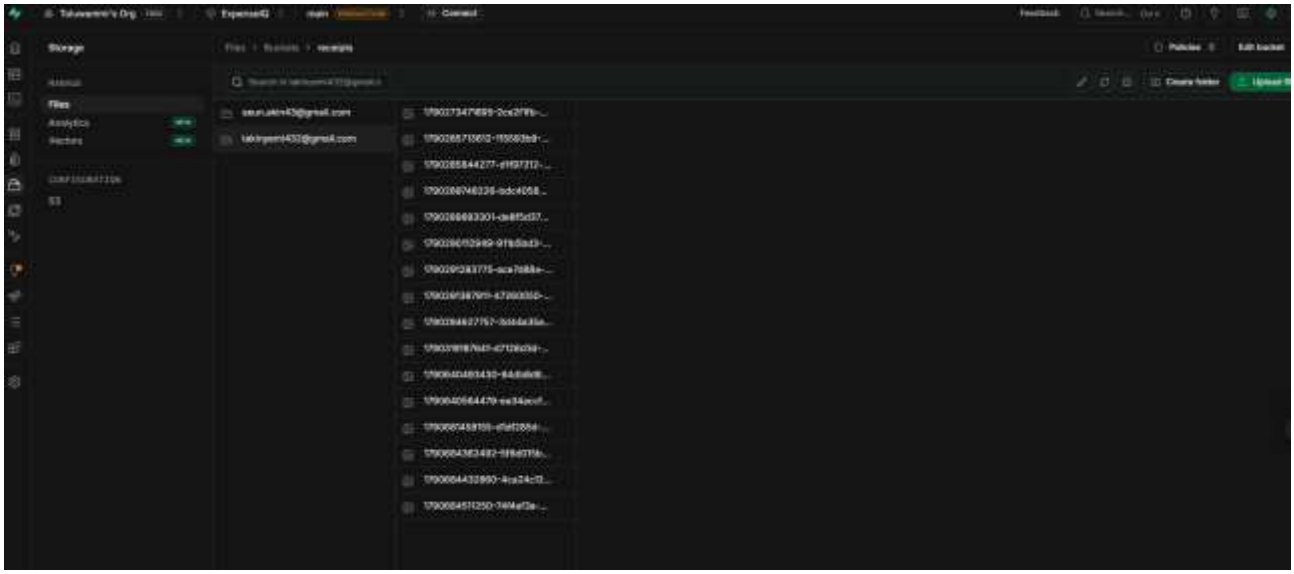


Figure 4: Supabase receipts storage bucket, with receipt files grouped in folders by employee email

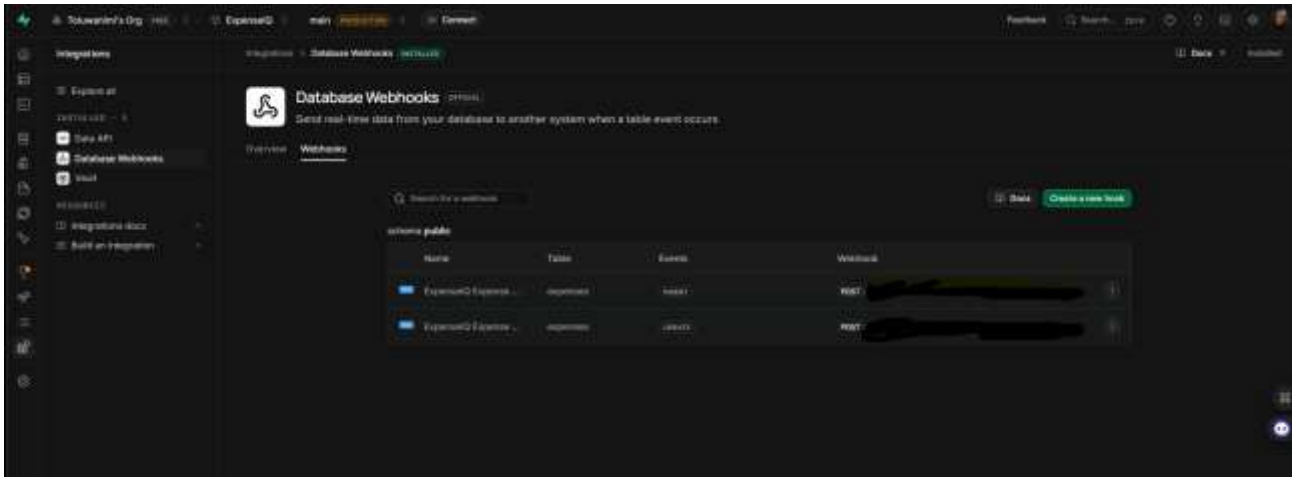


Figure 5: Supabase Database Webhooks: INSERT and UPDATE hooks on the expenses table (endpoint URLs redacted)

7. Lovable Frontend Implementation

The ExpenseIQ web application is live at https://expense-ai-pro-14.lovable.app/login. Employees and managers sign in from this page; accounts are created by an administrator.

Test login: username manager@test.com, password test@1234

7.1 Employee Dashboard

- Submit Expense
- My Expenses
- Profile

Users sign in with a work email and password (Figure 6). There is no self-registration: accounts are created by an administrator and linked to a role in the user_profiles table.

The submission form (Figure 7) pre-fills and locks the employee name and email from the signed-in profile. It captures department (dropdown), expense date, vendor/supplier, amount in NGN, purpose/business reason, payment method (Cash, Card, Bank Transfer or Personal Funds), optional project/client, receipt (PDF, JPG or PNG up to 10 MB) and optional notes.

If no receipt is attached, the form shows the warning "No receipt attached — this may delay processing" (Figure 8) but does not block submission; the record is then flagged with receipt_missing = true. The My Expenses view (Figure 9) lets employees track each expense by reference, vendor, amount, date, AI category and status.

7.2 Manager/Admin Dashboard

- Overview
- Pending Approvals
- All Expenses
- Reports

The Overview page (Figure 10) provides a date-range filter and summary cards for total expenses, total amount, Approved, Rejected, Awaiting approval and Needs clarification, together with a spend-by-department chart.

Pending Approvals (Figure 11) lists expenses requiring review. Each card shows the employee, department and expense reference, vendor, amount and date, the AI-suggested category (which the manager can change from a dropdown), a duplicate-risk badge, the AI reviewer note where one exists, a link to the receipt, and the Approve, Reject and Needs Clarification actions. Managers can also provide decision notes. The AI category remains a suggestion; the manager's decision and any category correction are the authoritative human input.

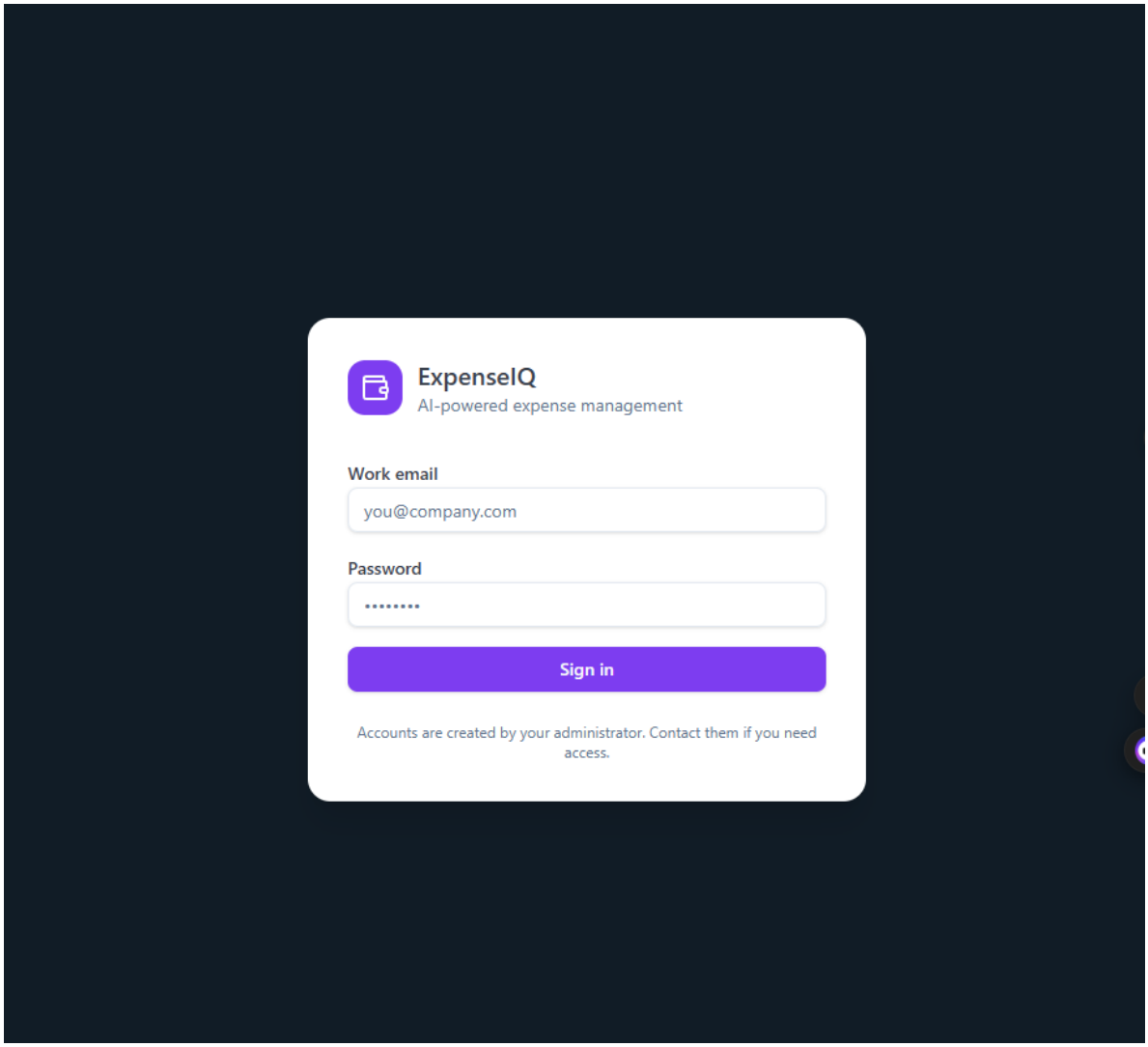


Figure 6: ExpenseIQ sign-in page; accounts are created by an administrator

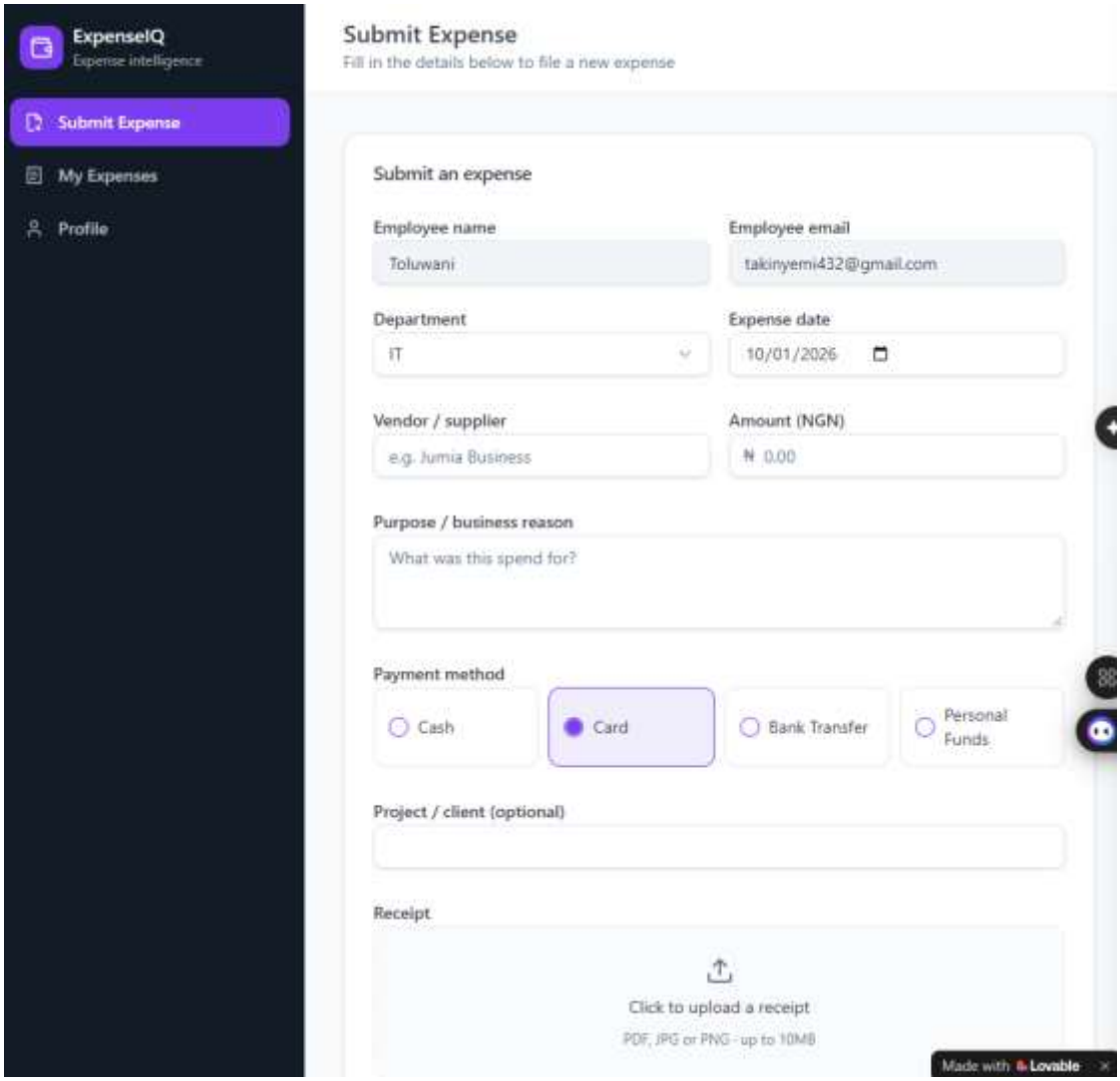


Figure 7: Employee Submit Expense form with pre-filled employee details, department, payment method and receipt upload

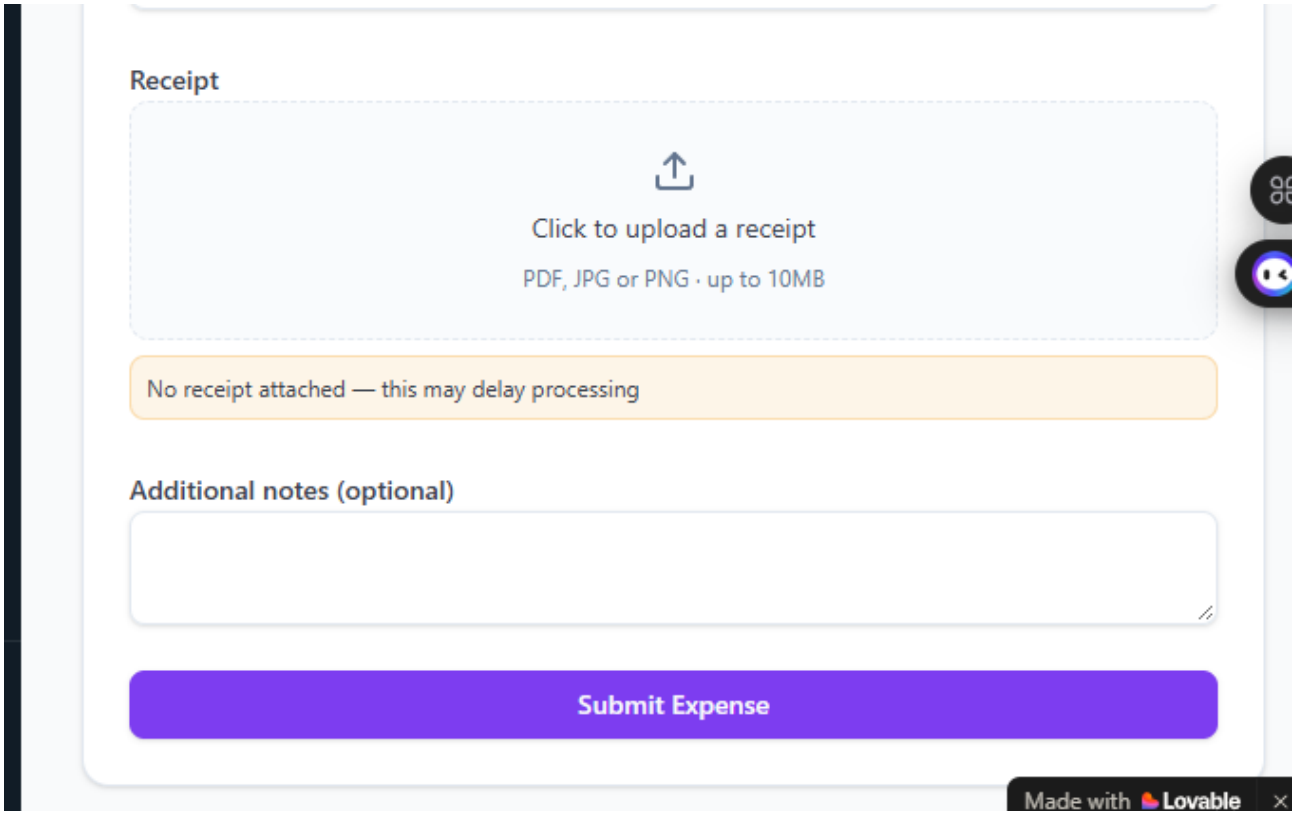


Figure 8: Missing-receipt warning on the Submit Expense form; submission is not blocked

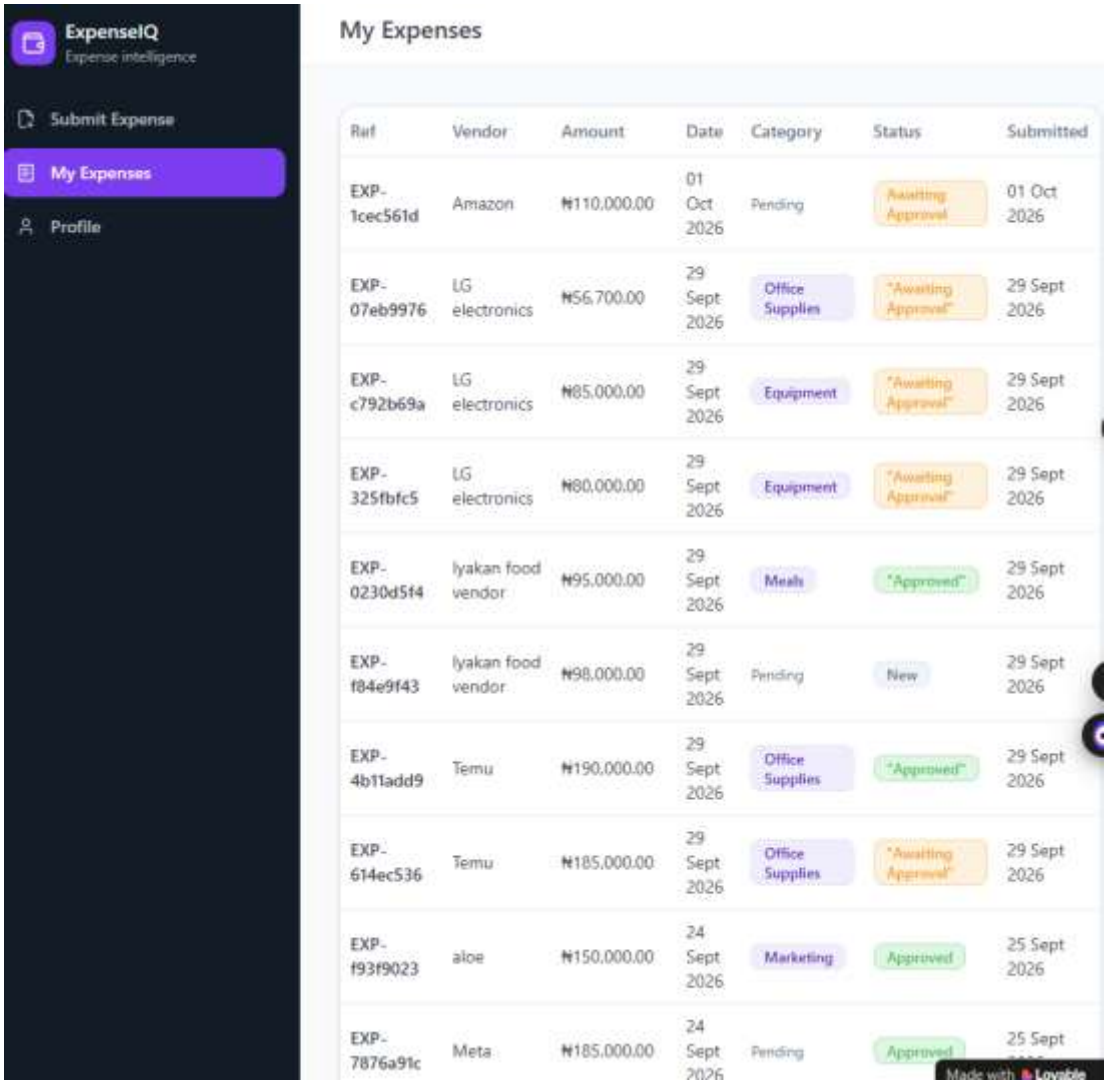


Figure 9: Employee My Expenses view showing reference, vendor, amount, date, AI category and status

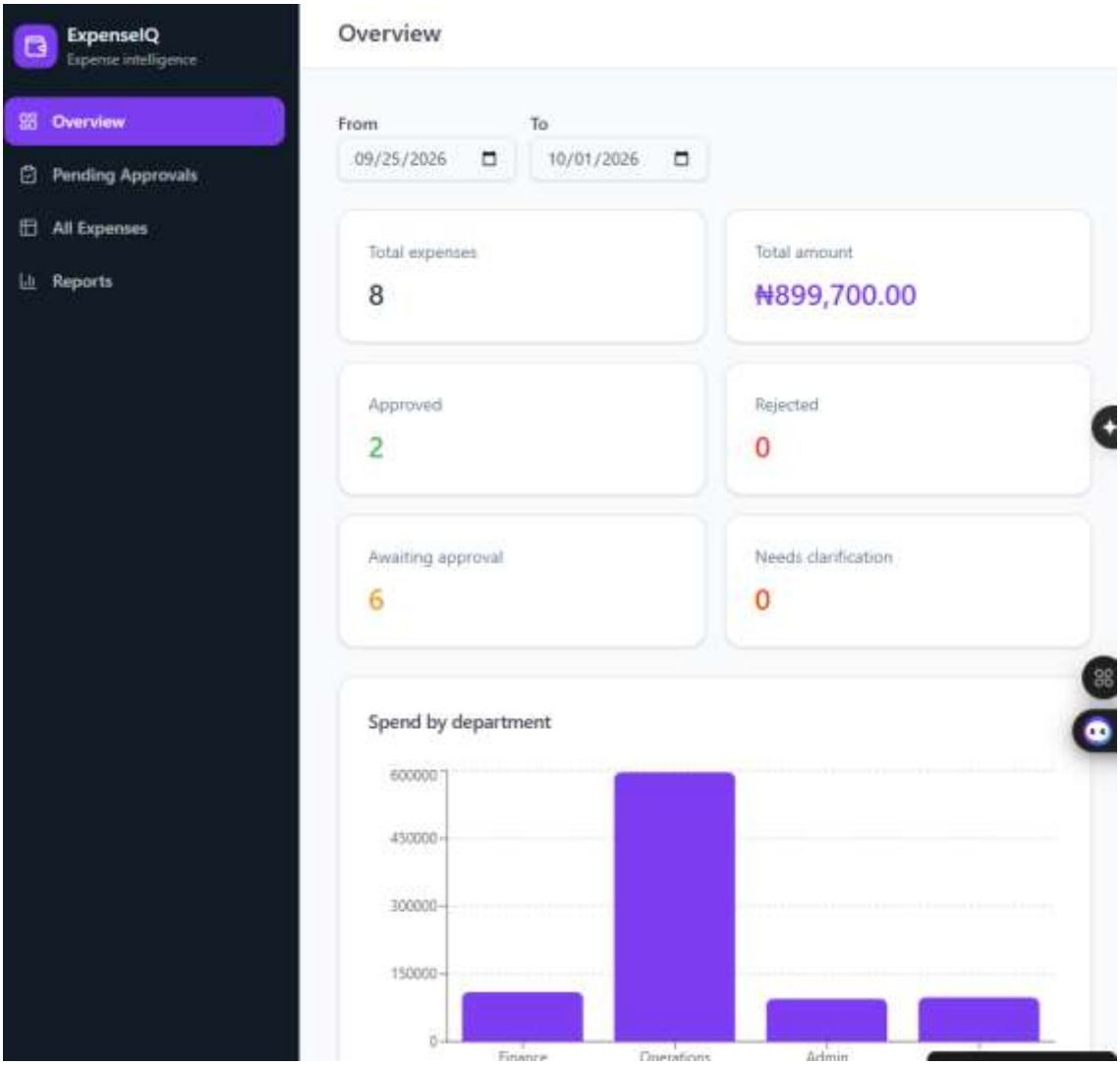


Figure 10: Manager Overview dashboard with date filter, status counts, total amount and spend by department

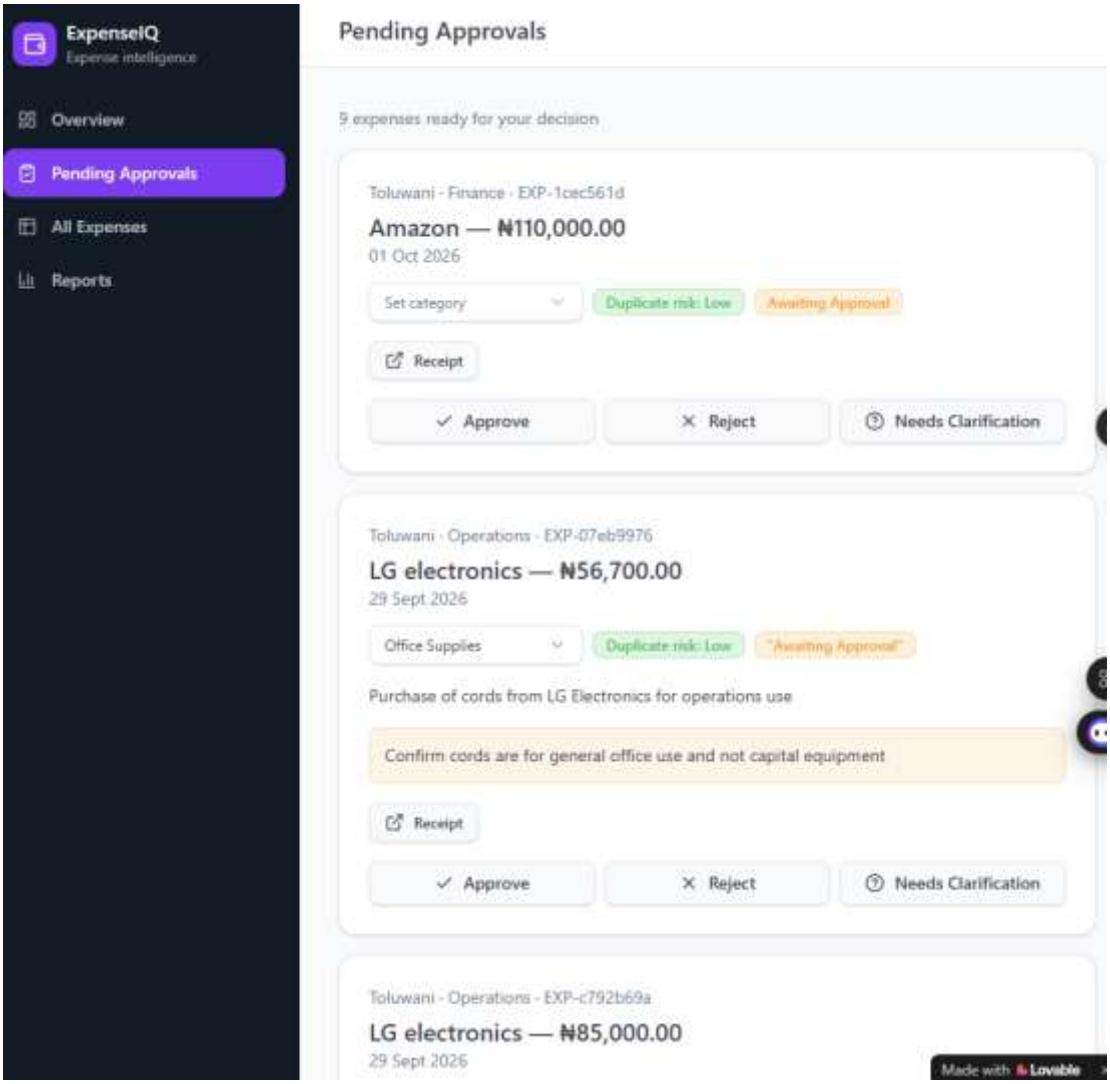


Figure 11: Manager Pending Approvals queue with category override, duplicate-risk badge, receipt link and decision actions

8. Make.com Scenario 1 — Expense Intake & AI Categorisation

Scenario 1 starts from a Supabase INSERT and processes the expense from submission through AI categorisation, storage, database enrichment, Airtable mirroring and notification routing.

Module	Function
1	Custom Webhook — receives the Supabase INSERT
2	Set Multiple Variables — extracts the record fields
3	HTTP GET — downloads the receipt
4	Google Drive — uploads the receipt
5	OpenRouter — AI categorisation
6	JSON — parses the AI output
7	Set Multiple Variables — AI and business-rule values
8	HTTP PATCH — updates Supabase
9	Airtable — creates the mirror record
10	Router — approval required vs auto-recorded
11A	Gmail — manager approval request
11B	Gmail — employee confirmation

The numbers in the module table show the order of execution. Inside Make.com each module also carries an internal ID: the module that extracts the record fields is referenced as 3 and the module holding the AI and business-rule values as 13, so mappings such as 13. ai_category or 3. amount in the screenshots refer to those internal IDs.

8.1 Fixed Categories

Marketing, Travel, Meals, Software, Office Supplies, Training, Professional Services, Logistics, Utilities, Equipment, Other

8.2 AI Output

The AI returns: category, subcategory, summary, duplicate_risk ("Low" or "Review"), and reviewer_note.

8.3 Receipt Flow

Lovable stores the submitted receipt in Supabase Storage. Make.com downloads the file and uploads it to Google Drive, then stores the durable Drive URL in Supabase.

Figure 12 shows the complete Scenario 1 canvas. Figure 13 shows the HTTP PATCH module (module 8), which writes the AI suggestions, the receipt URL, receipt_missing, approval_required and status back to the expense record in Supabase. The Airtable mirror created by module 9 is shown in Section 11 (Figure 23).

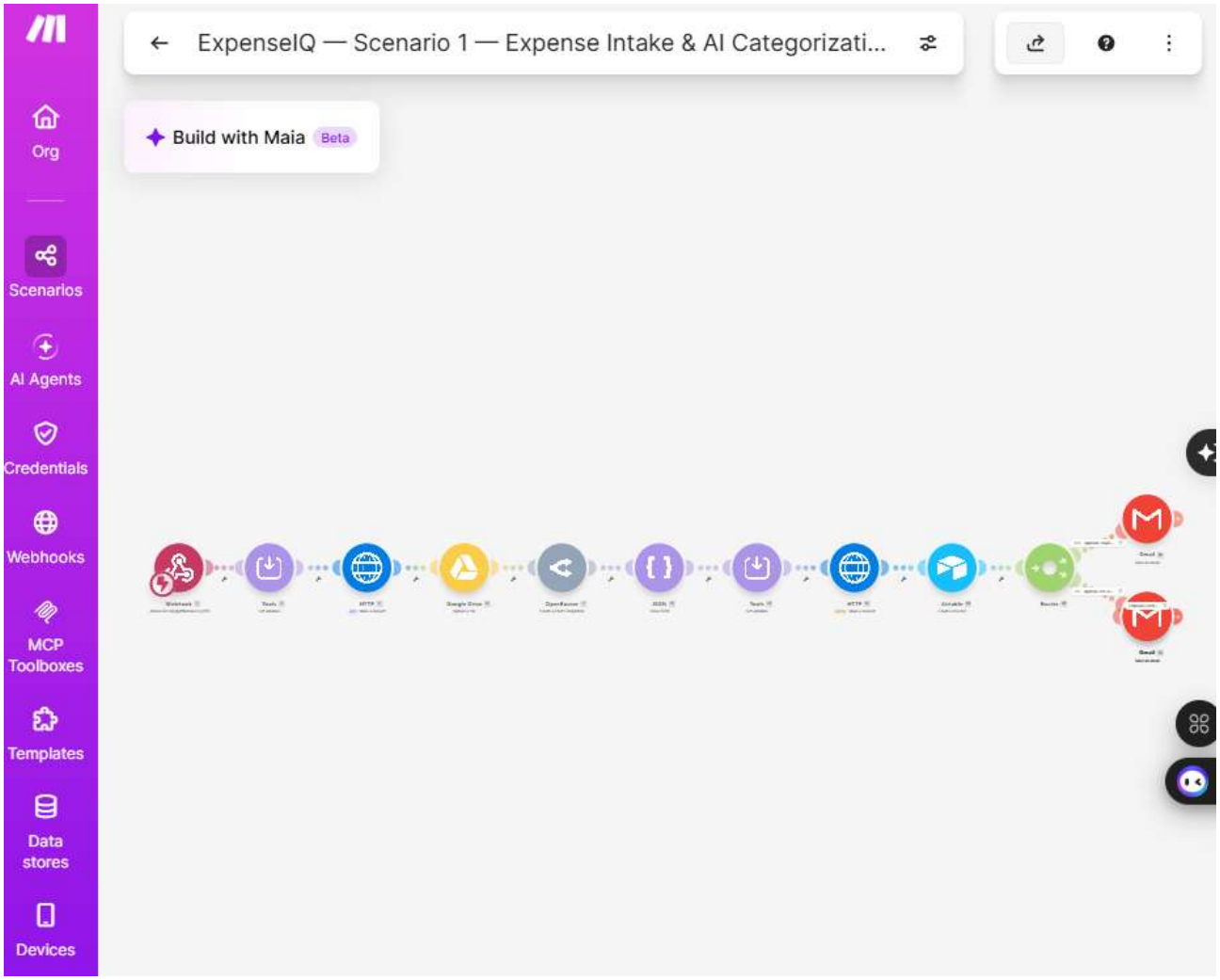


Figure 12: Scenario 1 full canvas in Make.com

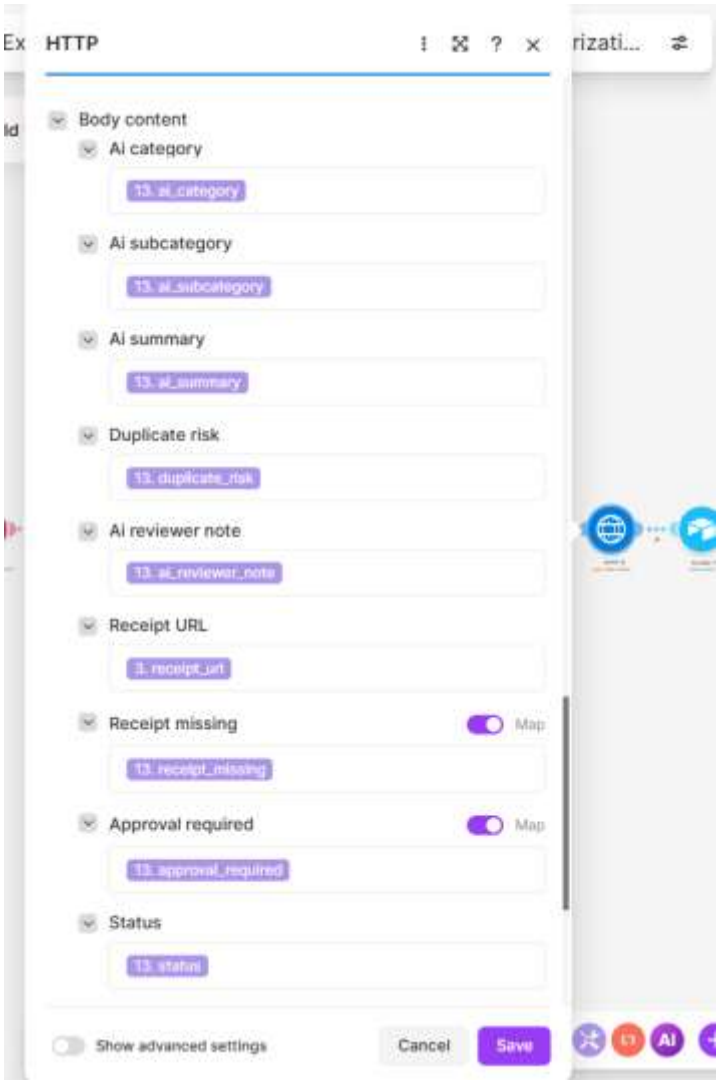


Figure 13: Scenario 1 HTTP PATCH module writing AI output, receipt fields, approval_required and status back to Supabase

8.4 Categorisation Prompt

The OpenRouter module uses the NVIDIA Nemotron 3 Ultra (free) model with JSON mode enabled and automatic fallback on. It sends the following system and user messages. The model is asked only for suggestion fields and must return valid JSON, which the JSON module then parses.

System message

```
You are an expense categorisation assistant for a Nigerian business.
Your job is to analyse the submitted expense and provide categorisation suggestions.
You must return ONLY valid JSON. You must NOT approve or reject expenses. You must NOT make
compliance or tax judgments.
Use only these categories: Marketing, Travel, Meals, Software, Office Supplies, Training,
Professional Services, Logistics, Utilities, Equipment, Other
```

User message

```
Analyse this business expense.
```

```
Department: {{department}}
Vendor: {{vendor}}
Amount: {{amount}}
Purpose: {{purpose}}
Payment Method: {{payment_method}}
Project/Client: {{project_client}}

Return only valid JSON using exactly this structure:
{ "category": "", "subcategory": "", "summary": "",
  "duplicate_risk": "Low or Review", "reviewer_note": "" }
```

The prompt requests a category, subcategory, summary, duplicate-risk flag and reviewer note only. It does not ask the model to approve, reject, judge tax or legal treatment, or label an expense as fraudulent; approval is decided by the explicit router rule in Section 9.

9. Approval Business Rules

Rule	Condition	Action
Manager approval	Amount ≥ NGN 100,000	Awaiting Approval + manager email
Auto-recorded	Amount < NGN 100,000	New + employee confirmation
Missing receipt	No receipt	receipt_missing = true; submission continues
Duplicate risk	AI returns Review	Review flag and reviewer note
Payment readiness	Status is not Approved	Not payment-ready; only Approved expenses may proceed to payment

The approval decision is an explicit Make.com business rule and is not determined by the AI. In Scenario 1 the rule is applied as two router filters on the numeric amount: the approval route requires amount ≥ 100000 (Figure 14) and the auto-recorded route requires amount < 100000 (Figure 15). The same rule is stored as the approval_required value written back to Supabase (Appendix A.2).

The manager approval request (Figure 16) summarises the expense, the AI suggestion and the reviewer note, and asks the manager to make the decision in ExpenseIQ. Employees whose expenses fall below the threshold receive a confirmation email (Figure 17).

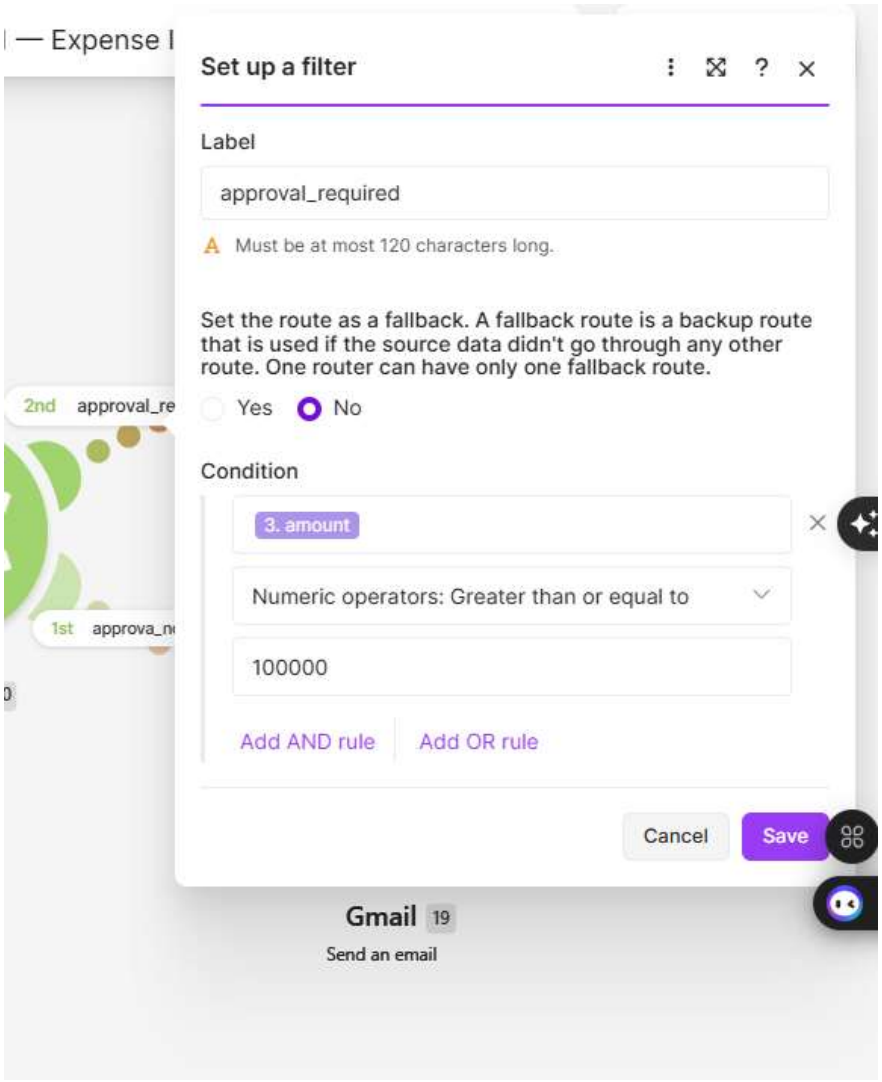


Figure 14: Router filter for the approval route: amount greater than or equal to 100000

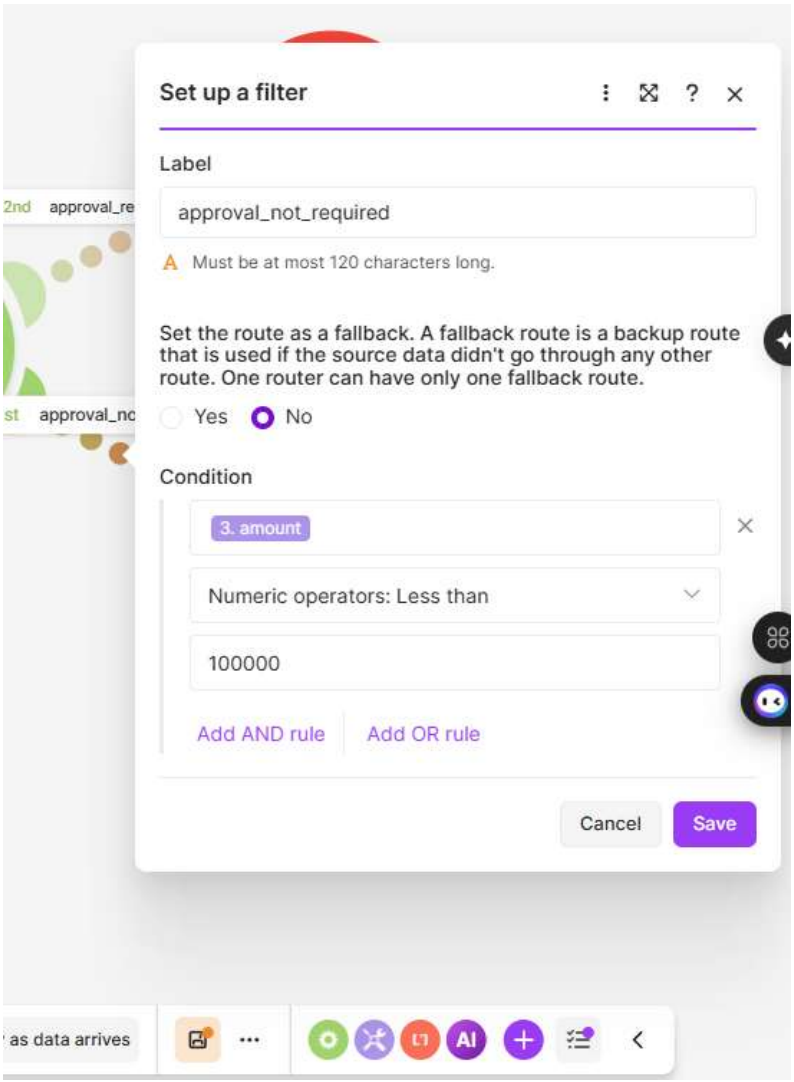
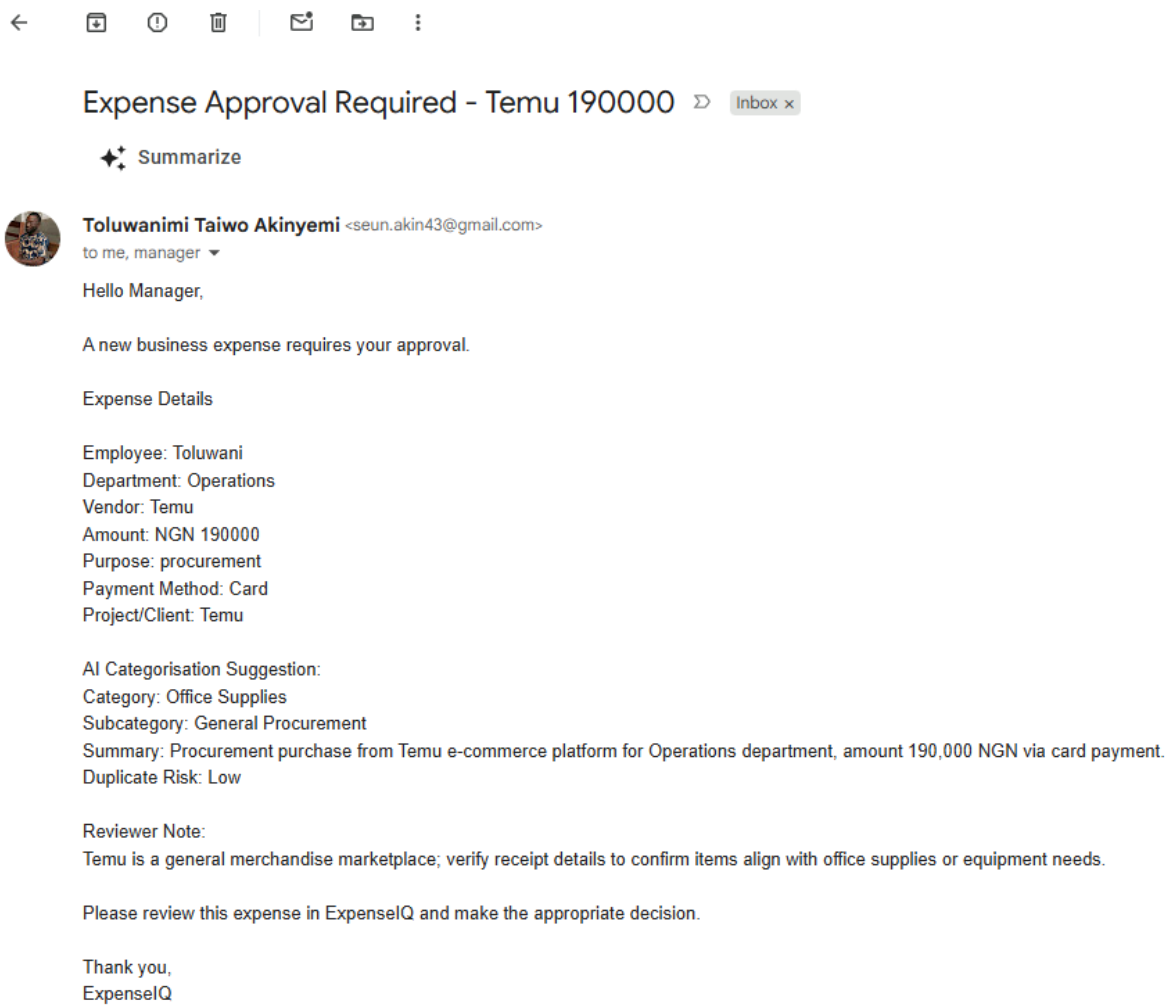


Figure 15: Router filter for the auto-recorded route: amount less than 100000



Expense Approval Required - Temu 190000 Inbox ×

Summarize

Toluwanimi Taiwo Akinyemi <seun.akin43@gmail.com>
to me, manager

Hello Manager,

A new business expense requires your approval.

Expense Details

Employee: Toluwani
Department: Operations
Vendor: Temu
Amount: NGN 190000
Purpose: procurement
Payment Method: Card
Project/Client: Temu

AI Categorisation Suggestion:
Category: Office Supplies
Subcategory: General Procurement
Summary: Procurement purchase from Temu e-commerce platform for Operations department, amount 190,000 NGN via card payment.
Duplicate Risk: Low

Reviewer Note:
Temu is a general merchandise marketplace; verify receipt details to confirm items align with office supplies or equipment needs.

Please review this expense in ExpenseIQ and make the appropriate decision.

Thank you,
ExpenseIQ

Figure 16: Manager approval request email for a NGN 190,000 Temu expense, including the AI suggestion and reviewer note

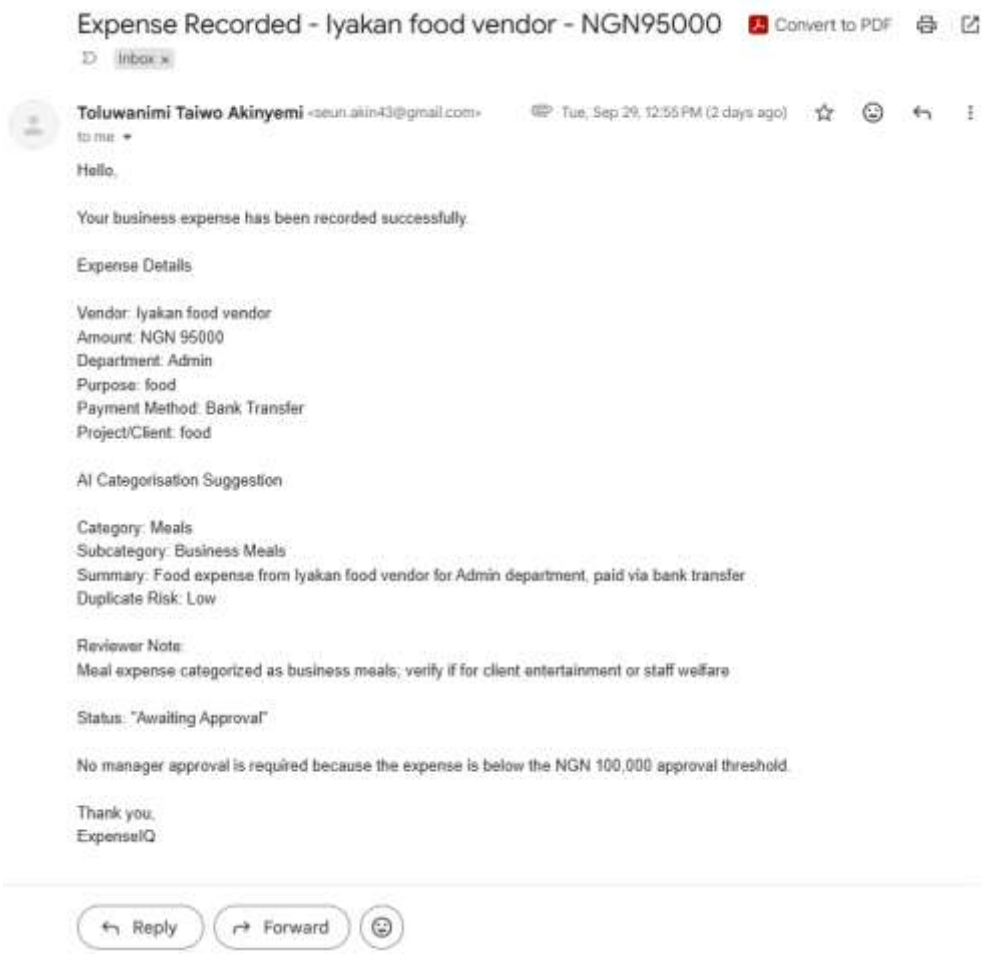


Figure 17: Employee confirmation email for a NGN 95,000 Iyakan food vendor expense below the approval threshold, showing the AI categorisation suggestion and reviewer note

10. Make.com Scenario 2 — Manager Decision & Employee Notification

Scenario 2 is triggered by an expense status update.

Stage	Function
1	Custom Webhook — receives the Supabase UPDATE
2	Notification Guard filter — notification_sent = false and old_record.status ≠ record.status
3	Router — Approved / Rejected / Needs Clarification
4A/B/C	Supabase PATCH — sets notification_sent = true
5A/B/C	Gmail — employee outcome notification

The notification_sent field is the primary duplicate-notification guard. The Notification Guard filter (Figure 19) lets an UPDATE webhook through only when notification_sent is false and the status has actually changed (old_record.status differs from record.status). Each PATCH module sets notification_sent to true, so the update made by the scenario itself is stopped by the filter and does not trigger a second email. The full scenario is shown in Figure 18.

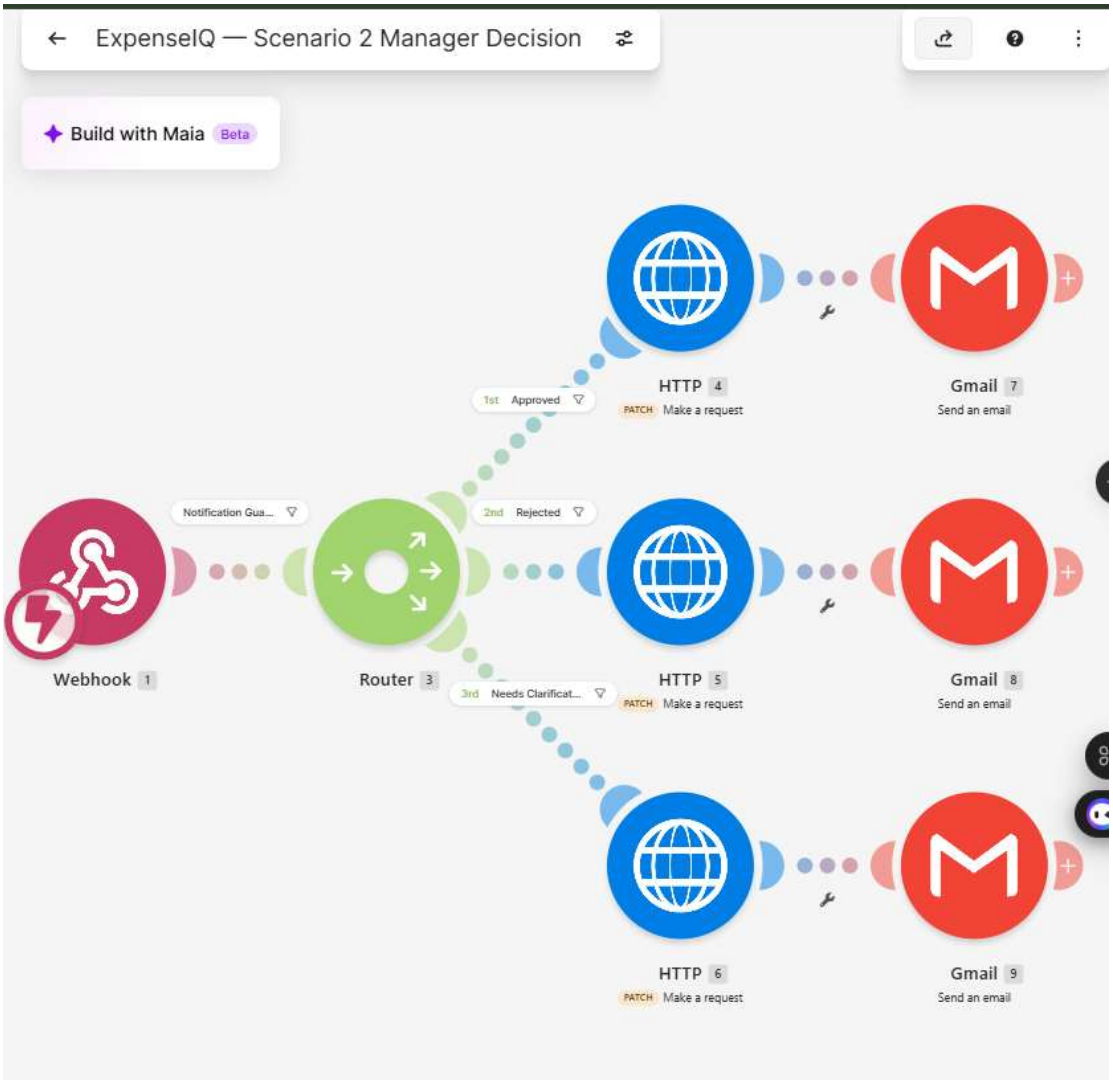


Figure 18: Scenario 2 full canvas: webhook, Notification Guard filter, router and the Approved, Rejected and Needs Clarification paths

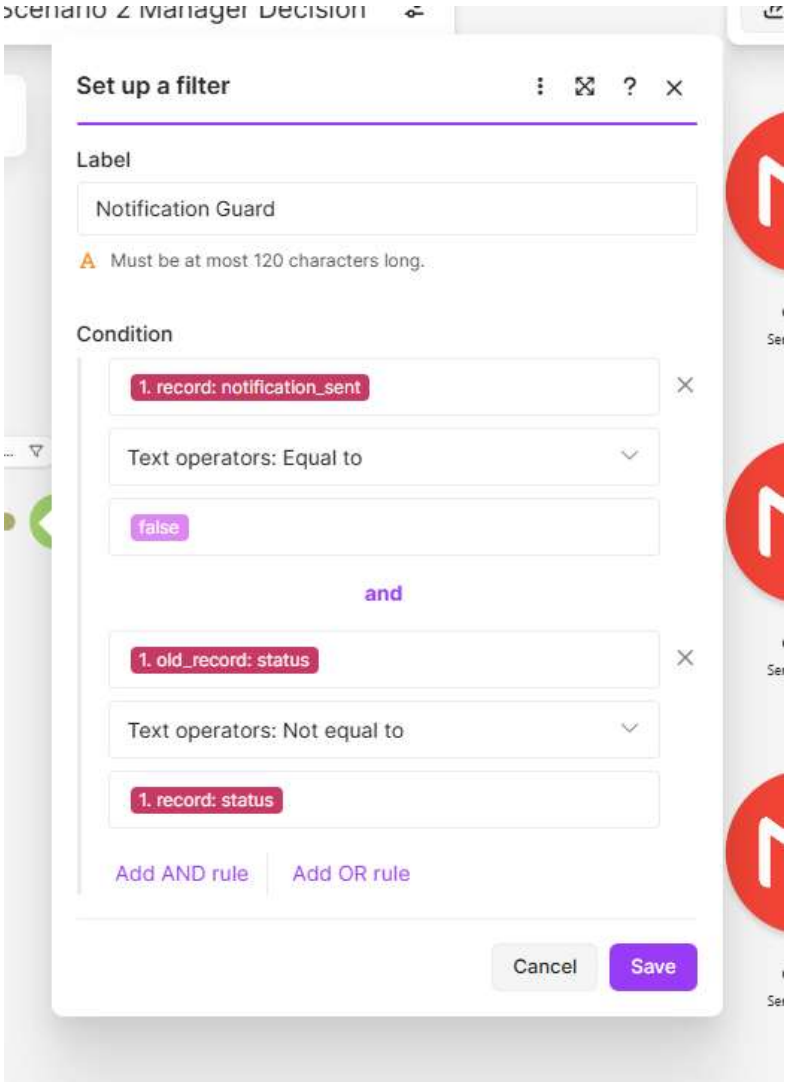


Figure 19: Notification Guard filter: notification_sent equals false and old_record.status differs from record.status

When the manager saves a decision, the employee receives one of three emails. Each email repeats the expense reference, vendor, amount, AI category and expense date, together with the manager notes. The colour of the notes panel follows the outcome: green for Approved (Figure 20), red for Rejected (Figure 21) and amber for Needs Clarification (Figure 22).

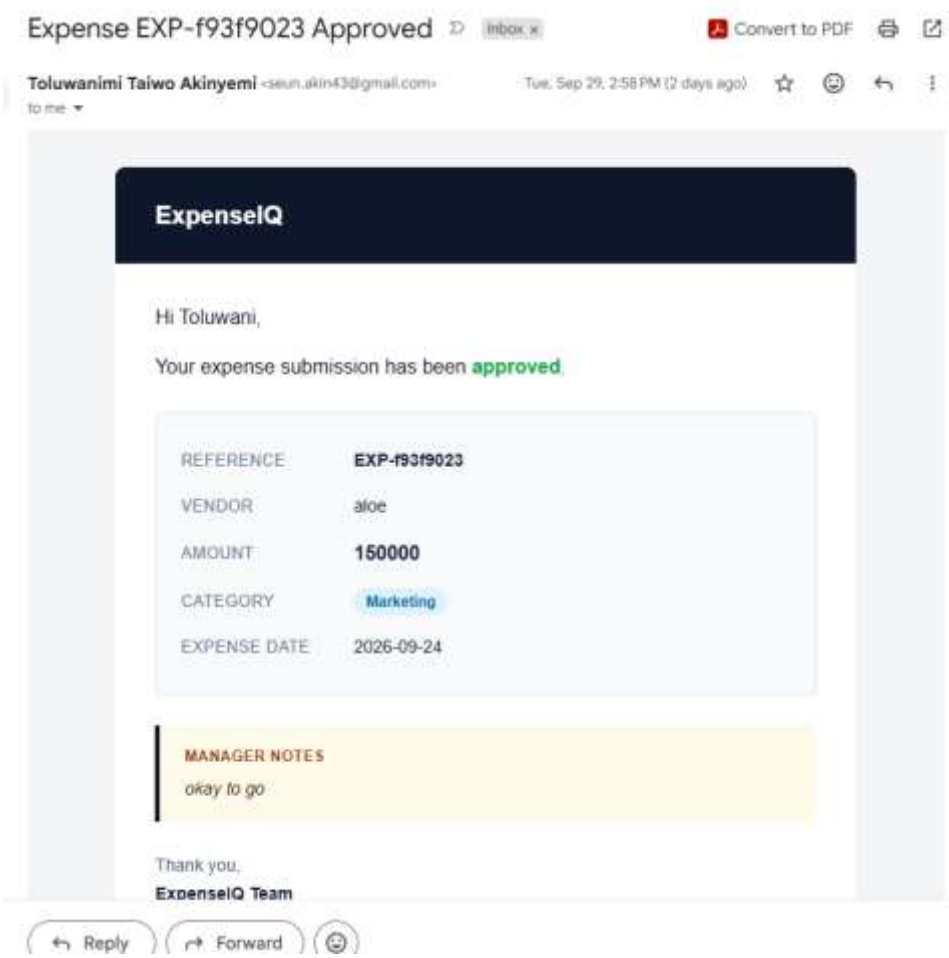


Figure 20: Approved outcome: employee email for expense EXP-f93f9023 (NGN 150,000, Marketing) with the manager notes

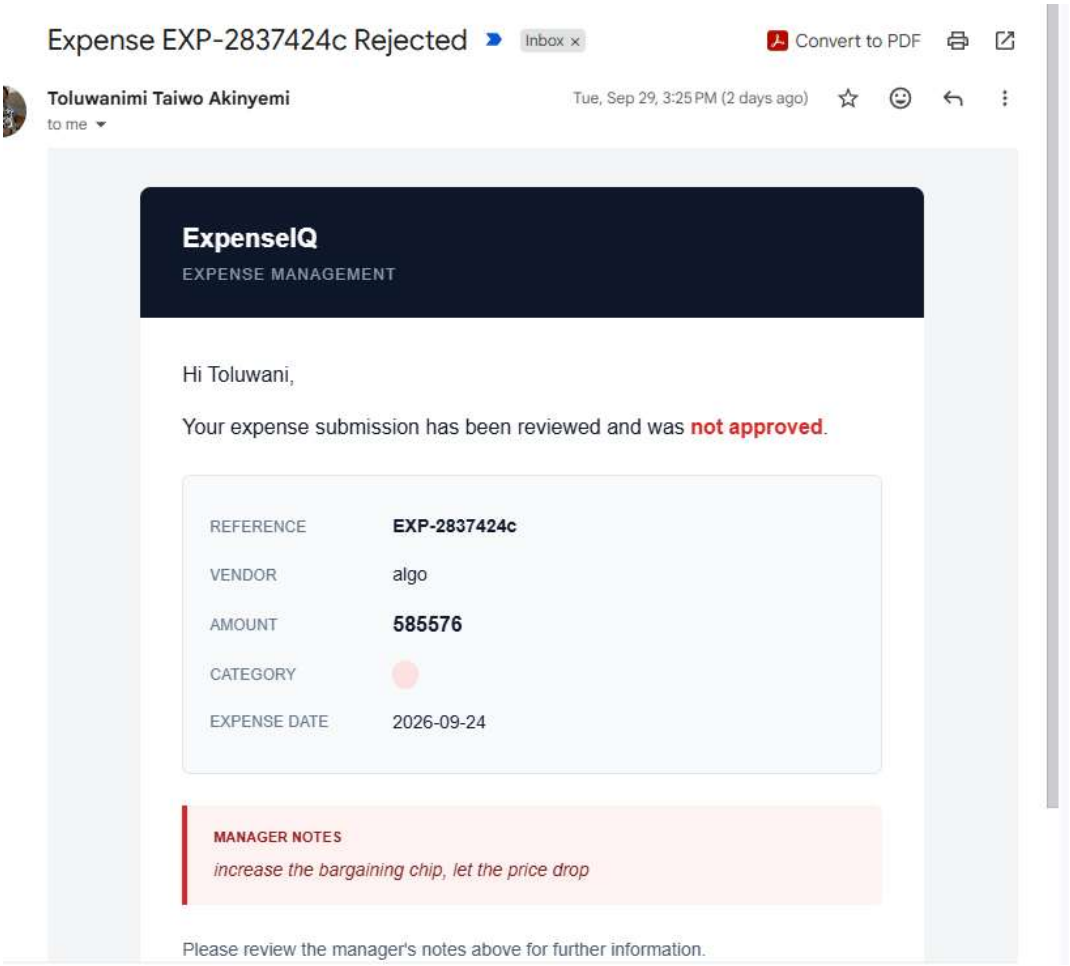


Figure 21: Rejected outcome: employee email for expense EXP-2837424c (NGN 585,576) with the manager notes

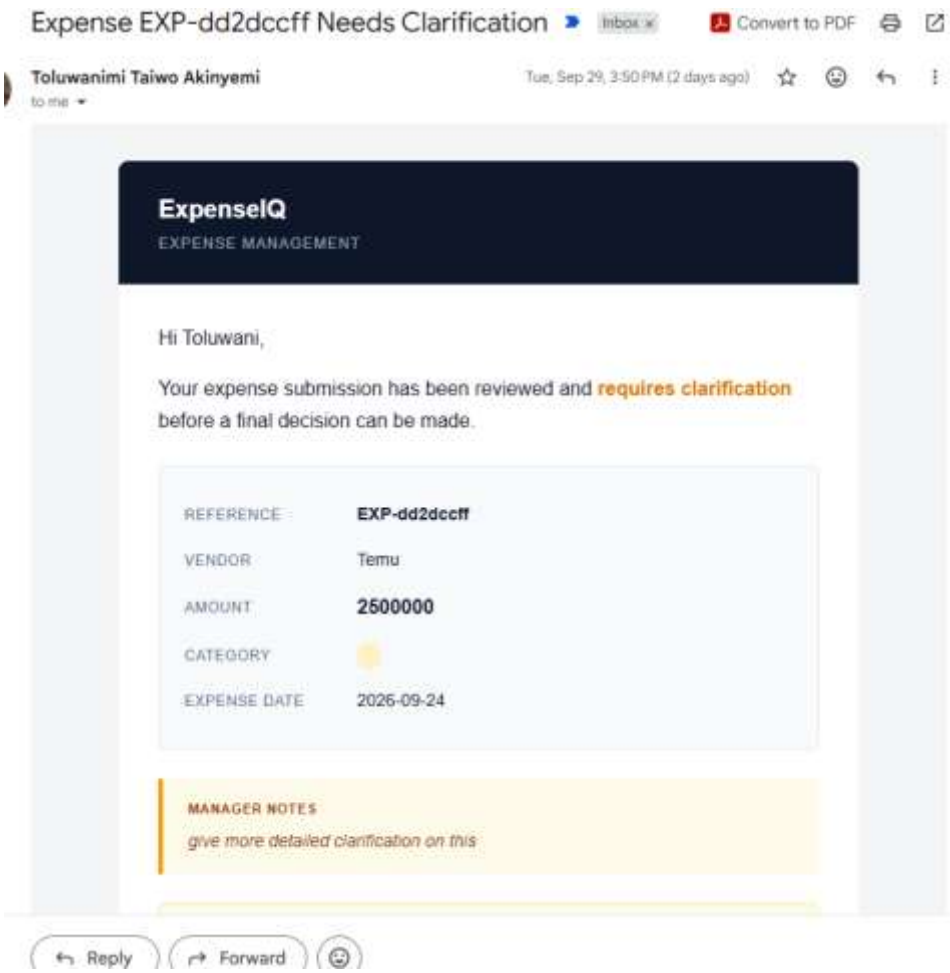


Figure 22: Needs Clarification outcome: employee email for expense EXP-dd2dccff (NGN 2,500,000) with the manager notes

11. Airtable Reporting Mirror

Airtable is a downstream management/reporting mirror. It does not initiate workflows. Make.com writes the processed expense to Airtable after Supabase is updated.

- Submission fields
- AI category/subcategory/summary
- Duplicate risk and reviewer note
- Approval required and status
- Manager decision information
- Receipt link

Each processed expense is written to the Expenses table in the ExpenseIQ Airtable base (Figure 23). The mirror holds the record_id from Supabase, the employee, vendor, amount, department, purpose, payment method and project/client, followed by the AI fields, the approval flags and the status. Repeated rows with the same record_id in the figure come from repeated test executions of Scenario 1; Supabase remains the single source of truth.

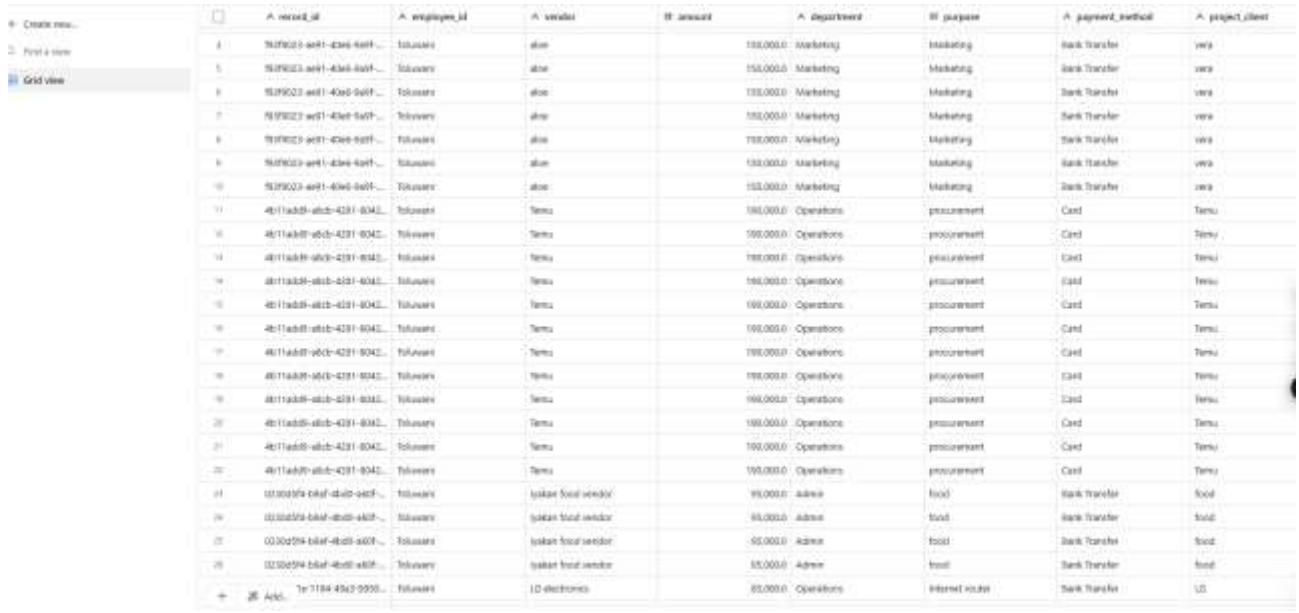


Figure 23: Airtable Expenses table mirroring processed expenses from Scenario 1

12. Make.com Scenario 3 — Weekly Summary Report

Scenario 3 is scheduled to run every Monday at 08:00 WAT (Africa/Lagos) and retrieves the expenses submitted in the previous seven days. The schedule is set on the scenario trigger; during testing the scenario was executed on demand.

Module	Function
1	HTTP GET — expenses from the last seven days from Supabase
2	Iterator — splits the returned records
3	Set Multiple Variables — week dates and report label
4	Array Aggregator — collects amount, status, department and AI category
5	Set Multiple Variables — totals and status metrics
6	Google Sheets — adds a row to Weekly Summary
7	Gmail — management report email
8	Iterator — iterates the aggregated records
9	Router — Category route / Department route
10	Numeric Aggregator — SUM by AI category
11	Google Sheets — adds rows to By Category
12	Numeric Aggregator — SUM by department
13	Google Sheets — adds rows to By Department

Weekly metrics include total expenses, grand total, approved, rejected, awaiting approval and needs clarification. The canvas is shown in Figure 24. The Google Sheets workbook has three tabs: By Category (Figure 25), By Department (Figure 26) and Weekly Summary (Figure 27). The management email (Figure 28) reports the same headline metrics and points readers to the workbook for the category and department detail. The sheets contain rows from several successive test runs.



Figure 24: Scenario 3 full canvas in Make.com

Weekly Summary - EXPENSEIQ

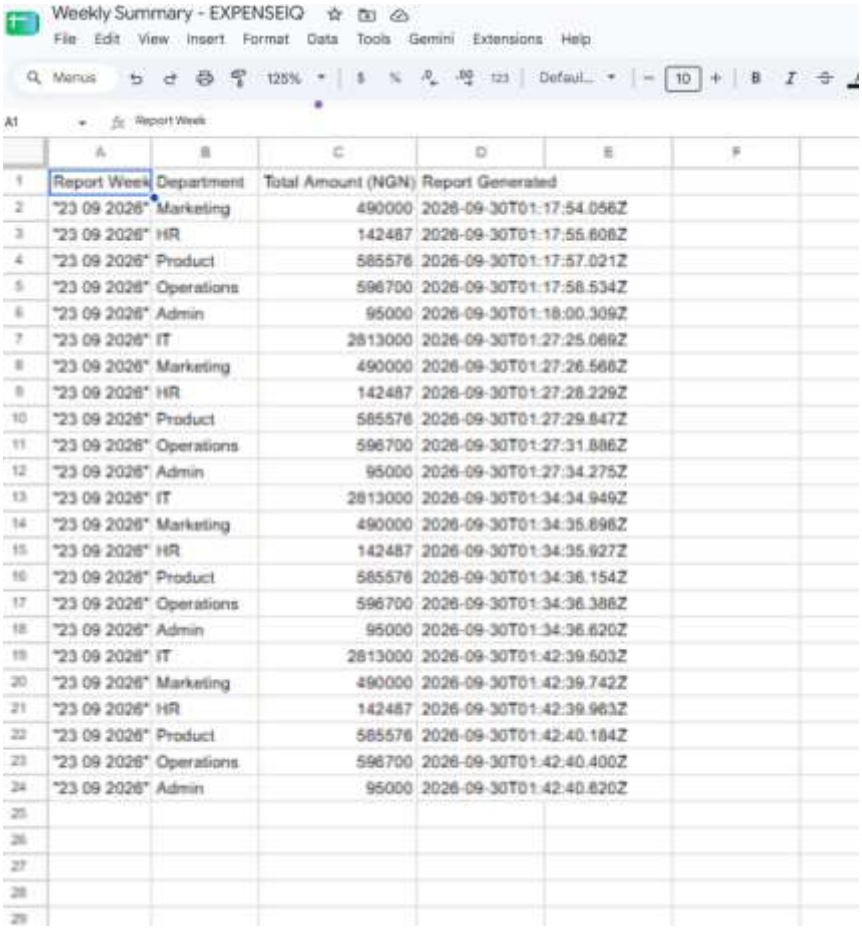
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A1 Report Week

	A	B	C	D	E
1	Report Week	Category	Total Amount (NGN)	Report Generatted	
2	"23 09 2026"		3881063	2026-09-30T01:17:44.961Z	
3	"23 09 2026"	Marketing	150000	2026-09-30T01:17:46.439Z	
4	"23 09 2026"	Office Sup	431700	2026-09-30T01:17:48.086Z	
5	"23 09 2026"	Meals	95000	2026-09-30T01:17:49.681Z	
6	"23 09 2026"	Equipmer	165000	2026-09-30T01:17:51.070Z	
7	"23 09 2026"		3881063	2026-09-30T01:27:12.969Z	
8	"23 09 2026"	Marketing	150000	2026-09-30T01:27:16.304Z	
9	"23 09 2026"	Office Sup	431700	2026-09-30T01:27:17.880Z	
10	"23 09 2026"	Meals	95000	2026-09-30T01:27:20.361Z	
11	"23 09 2026"	Equipmer	165000	2026-09-30T01:27:22.678Z	
12	"23 09 2026"		3881063	2026-09-30T01:34:33.751Z	
13	"23 09 2026"	Marketing	150000	2026-09-30T01:34:34.013Z	
14	"23 09 2026"	Office Sup	431700	2026-09-30T01:34:34.245Z	
15	"23 09 2026"	Meals	95000	2026-09-30T01:34:34.478Z	
16	"23 09 2026"	Equipmer	165000	2026-09-30T01:34:34.708Z	
17	"23 09 2026"		3881063	2026-09-30T01:42:38.364Z	
18	"23 09 2026"	Marketing	150000	2026-09-30T01:42:38.603Z	
19	"23 09 2026"	Office Sup	431700	2026-09-30T01:42:38.830Z	
20	"23 09 2026"	Meals	95000	2026-09-30T01:42:39.052Z	
21	"23 09 2026"	Equipmer	165000	2026-09-30T01:42:39.278Z	
22					
23					

Figure 25: By Category sheet: weekly totals per AI category



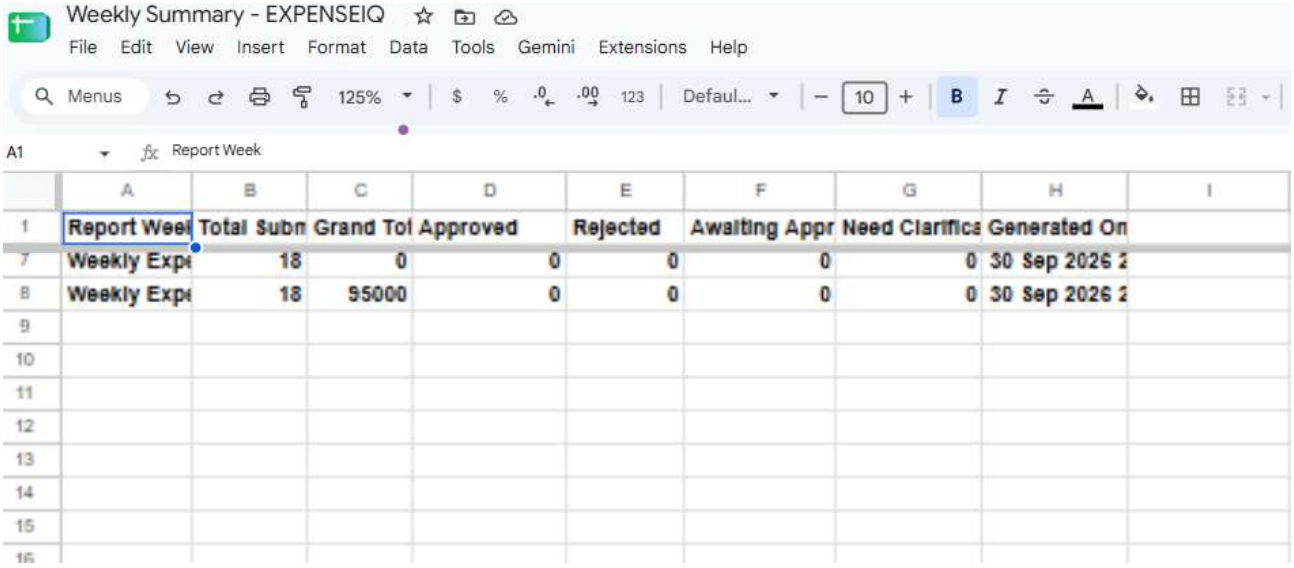
Weekly Summary - EXPENSEIQ

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A1 Report Week

	A	B	C	D	E	F
1	Report Week	Department	Total Amount (NGN)	Report Generated		
2	"23 09 2026"	Marketing	490000	2026-09-30T01:17:54.056Z		
3	"23 09 2026"	HR	142487	2026-09-30T01:17:55.608Z		
4	"23 09 2026"	Product	585576	2026-09-30T01:17:57.021Z		
5	"23 09 2026"	Operations	596700	2026-09-30T01:17:58.534Z		
6	"23 09 2026"	Admin	95000	2026-09-30T01:18:00.309Z		
7	"23 09 2026"	IT	2813000	2026-09-30T01:27:25.069Z		
8	"23 09 2026"	Marketing	490000	2026-09-30T01:27:26.566Z		
9	"23 09 2026"	HR	142487	2026-09-30T01:27:28.229Z		
10	"23 09 2026"	Product	585576	2026-09-30T01:27:29.847Z		
11	"23 09 2026"	Operations	596700	2026-09-30T01:27:31.886Z		
12	"23 09 2026"	Admin	95000	2026-09-30T01:27:34.275Z		
13	"23 09 2026"	IT	2813000	2026-09-30T01:34:34.949Z		
14	"23 09 2026"	Marketing	490000	2026-09-30T01:34:35.696Z		
15	"23 09 2026"	HR	142487	2026-09-30T01:34:35.927Z		
16	"23 09 2026"	Product	585576	2026-09-30T01:34:36.154Z		
17	"23 09 2026"	Operations	596700	2026-09-30T01:34:36.386Z		
18	"23 09 2026"	Admin	95000	2026-09-30T01:34:36.620Z		
19	"23 09 2026"	IT	2813000	2026-09-30T01:42:39.503Z		
20	"23 09 2026"	Marketing	490000	2026-09-30T01:42:39.742Z		
21	"23 09 2026"	HR	142487	2026-09-30T01:42:39.963Z		
22	"23 09 2026"	Product	585576	2026-09-30T01:42:40.184Z		
23	"23 09 2026"	Operations	596700	2026-09-30T01:42:40.400Z		
24	"23 09 2026"	Admin	95000	2026-09-30T01:42:40.620Z		
25						
26						
27						
28						
29						

Figure 26: By Department sheet: weekly totals per department



Weekly Summary - EXPENSEIQ

File Edit View Insert Format Data Tools Gemini Extensions Help

A1 Report Week

	A	B	C	D	E	F	G	H	I
1	Report Weel	Total Subn	Grand Tol	Approved	Rejected	Awaiting Appr	Need Clarifica	Generated On	
7	Weekly Expe	18	0	0	0	0	0	30 Sep 2026 2	
8	Weekly Expe	18	95000	0	0	0	0	30 Sep 2026 2	
9									
10									
11									
12									
13									
14									
15									
16									

Figure 27: Weekly Summary sheet: report label, submission count, totals, status counts and generation time

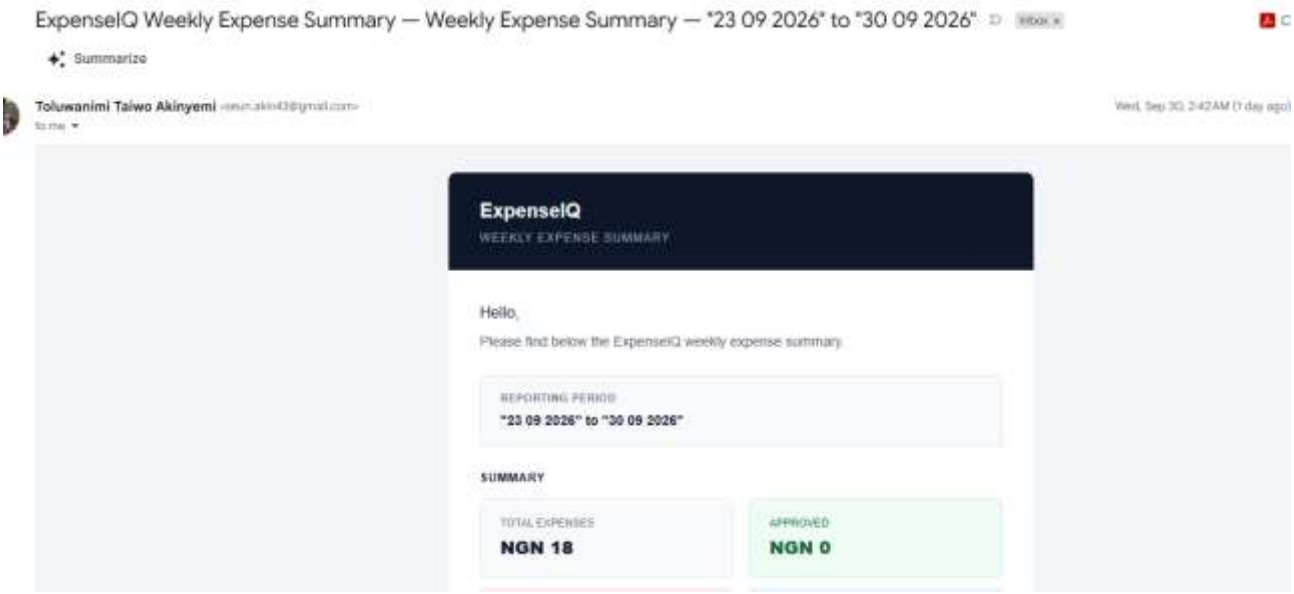


Figure 28: Weekly management email sent by Scenario 3

13. Testing and Results

13.1 Test Cases and Results

The test cases below were run against the live system using expenses submitted through the Lovable interface. Each result is supported by the figure shown in the last column.

ID	Test and input	Expected result	Result and evidence
TC01	Approval required: Temu, Operations, NGN 190,000	Routed for approval; manager email sent	Passed. Manager approval email received (Figures 14 and 16)
TC02	Auto-recorded: Iyakan food vendor, Admin, NGN 95,000	Below threshold; employee confirmation, no manager approval	Passed. Confirmation email received (Figures 15 and 17)
TC03	AI categorisation of the Iyakan food vendor expense	Category from the fixed list, subcategory, summary, duplicate risk and reviewer note	Passed. Meals / Business Meals, duplicate risk Low (Figure 17)
TC04	Write-back to Supabase after AI processing	AI fields, receipt URL, approval_required and status saved	Passed. PATCH module configured with all fields (Figure 13)
TC05	Missing receipt: submit the form without a file	Warning shown; submission not blocked	Passed. Warning displayed on the form (Figure 8)
TC06	Manager Approve: aloe, Marketing, NGN 150,000	Status Approved; approval email with manager notes	Passed (Figure 20)
TC07	Manager Reject: algo, Product, NGN 585,576	Status Rejected; rejection email with manager notes	Passed (Figure 21)
TC08	Needs Clarification: Temu, NGN 2,500,000	Needs Clarification; email with manager notes	Passed (Figure 22)
TC09	Duplicate-notification guard	Only a real status change sends an email; the scenario's own update does not re-trigger it	Verified by configuration (Figure 19)
TC10	Airtable mirror	Processed expenses appear in Airtable	Passed (Figure 23)
TC11	Weekly report: run Scenario 3	Three sheet tabs populated; management email sent	Passed (Figures 24 to 28)
TC12	Scenario deployment	Scenarios 1 and 2 active; Scenario 3 available for scheduled use	Passed (Figure 29)

All twelve test cases produced the expected routing and notifications. Together they cover submission, AI categorisation, the approval threshold, the three manager decisions, employee notification, the Airtable mirror and the weekly report.

13.2 Scenario Deployment Status

Figure 29 shows the Make.com scenario list. Scenarios 1 and 2 are switched on and respond to Supabase webhooks. Scenario 3 is switched off at the time of capture; it is switched on to run on its Monday schedule.

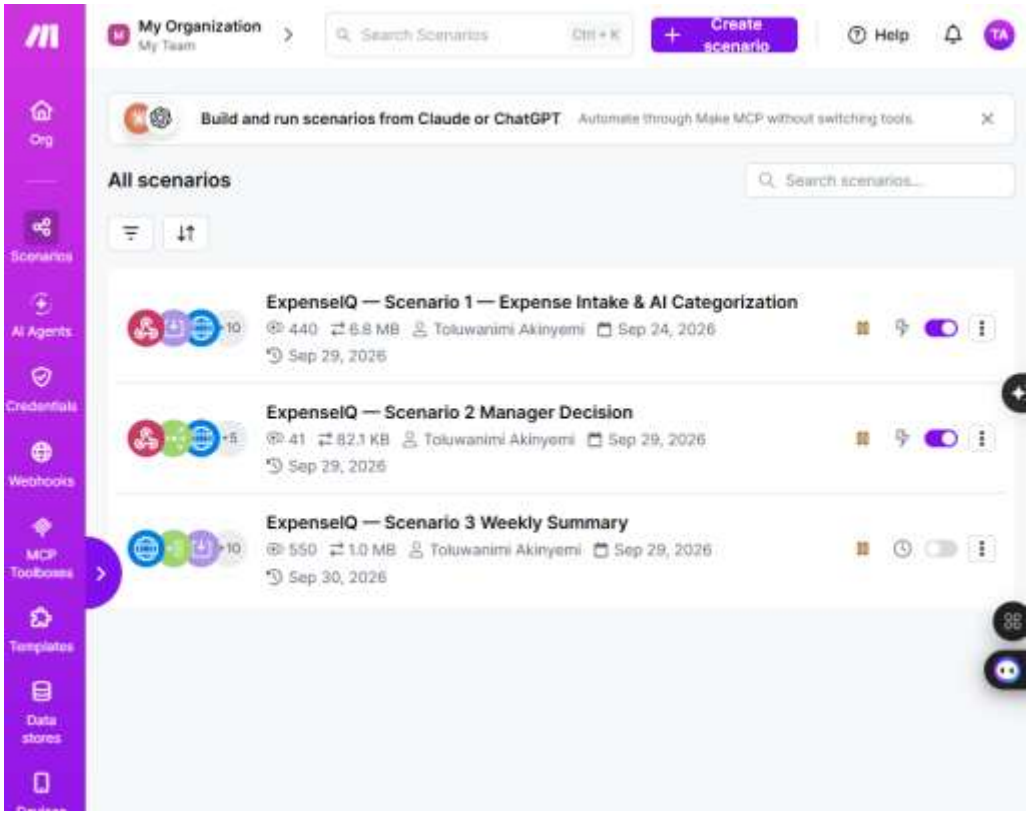


Figure 29: Make.com scenarios dashboard showing the three ExpenseIQ scenarios

14. Evidence Register

The table lists every figure in this report, the purpose it serves and the section in which it appears. Webhook endpoint URLs are redacted and no credentials or API keys are shown in the screenshots.

Figure	Evidence	Purpose	Section
1	System architecture diagram	Architecture overview	4
2	Supabase expenses table	Database evidence	6.5
3	Supabase user profiles	Role evidence	6.5
4	Supabase receipts storage	Receipt evidence	6.5
5	Supabase Database Webhooks	Trigger evidence	6.5
6	Lovable login page	Access control evidence	7.2
7	Lovable Submit Expense form	Submission evidence	7.2
8	Missing-receipt warning	Submission evidence	7.2
9	Employee My Expenses view	Employee dashboard evidence	7.2
10	Manager Overview dashboard	Manager dashboard evidence	7.2
11	Manager Pending Approvals	Approval queue evidence	7.2
12	Scenario 1 full canvas	Automation evidence	8.3
13	Scenario 1 Supabase PATCH	Database update evidence	8.3
14	Approval route filter (amount ≥ 100000)	Threshold evidence	9
15	Auto-recorded route filter (amount < 100000)	Threshold evidence	9
16	Manager approval email	Notification evidence	9
17	Employee confirmation email	Notification evidence	9

Figure	Evidence	Purpose	Section
18	Scenario 2 full canvas	Decision automation evidence	10
19	Scenario 2 Notification Guard	De-duplication evidence	10
20	Approved outcome email	Decision evidence	10
21	Rejected outcome email	Decision evidence	10
22	Needs Clarification outcome email	Decision evidence	10
23	Airtable expense mirror	Reporting mirror evidence	11
24	Scenario 3 full canvas	Weekly automation evidence	12
25	By Category sheet	Category aggregation evidence	12
26	By Department sheet	Department aggregation evidence	12
27	Weekly Summary sheet	Overall reporting evidence	12
28	Weekly management email	Weekly notification evidence	12
29	Make.com scenarios dashboard	Overall system evidence	13.2

15. AI Role Boundaries

The AI only:

- Suggests a category from the fixed list.
- Suggests a subcategory.
- Writes a concise summary.
- Flags duplicate risk as Low or Review.
- Writes a reviewer note for unclear information.

The AI does not:

- Approve or reject expenses.
- Make payment decisions.
- Give tax or legal conclusions.
- Override the approval threshold.
- Act as the final financial decision-maker.

16. Security and Data Protection Considerations

- Supabase is the authoritative database and RLS is enabled.
- The frontend uses the anon/public key; the service-role key is restricted to Make.com server-side HTTP modules.
- Service-role keys, API keys, passwords and webhook secrets must not appear in documentation screenshots.
- The receipts bucket is private.
- Database Webhooks trigger Make.com workflows.
- notification_sent helps prevent duplicate employee notifications.
- Original submitted values are preserved while decision/workflow fields are appended.

17. Expected Outcomes

Area	Outcome

Submission	Employee expense records are created in Supabase
AI	Structured categorisation and review suggestions are produced
Approval	NGN 100,000 threshold routes expenses correctly
Receipts	Receipts are stored durably in Google Drive
Mirror	Processed records appear in Airtable
Notifications	Managers/employees receive appropriate emails
Decision workflow	Approved, Rejected and Needs Clarification paths are supported
Reporting	Weekly, category and department summaries are written to Google Sheets

18. Limitations and Future Improvements

- Add a detailed immutable decision-history table for more extensive auditing.
- Add a Paid status and a payment-ready queue so that only Approved expenses can proceed to payment.
- Enhance duplicate detection using historical expense similarity.
- Support configurable thresholds if future business requirements need them.
- Add richer dashboards and downloadable management reports.
- Add stronger retry/error reporting across external services.
- Introduce automated regression tests for Make.com scenarios.
- Expand role and permission controls if additional organizational roles are introduced.
- Apply the NGN 100,000 comparison directly to the numeric amount in the Scenario 1 variables module, as the router filters already do, so that the status label and approval_required value saved to Supabase and shown in confirmation emails always match the router decision.
- Add an Airtable search-then-update step keyed on record_id so that repeated executions of Scenario 1 update one mirror row instead of adding a new one.
- Review the Scenario 3 count and sum expressions so that the grand total and the approved, rejected, awaiting and clarification counts are calculated from the aggregated array, and label counts as numbers rather than NGN amounts.

19. Conclusion

ExpenseIQ demonstrates how AI, no-code automation and a modern web application can be combined to automate a structured business expense lifecycle. Supabase provides the authoritative record, Make.com performs explicit workflow and business-rule processing, and AI provides controlled categorisation and review assistance.

The system covers the complete lifecycle from employee submission through AI-assisted processing, receipt storage, approval routing, human decision, employee notification and weekly management reporting. The separation between AI suggestions and human financial decisions is maintained throughout the design.

Testing confirmed the main paths of the design: threshold routing, AI categorisation, receipt handling, the three manager decisions with employee notification, the Airtable mirror and the weekly management report. The remaining improvements listed in Section 18 are refinements to expressions and controls rather than changes to the architecture.

Appendix A — Key Configuration Reference

A.1 Fixed Categories

Marketing, Travel, Meals, Software, Office Supplies, Training, Professional Services, Logistics, Utilities, Equipment, Other

A.2 Approval Formula

approval_required = if(parseNumber(amount) >= 100000; true; false)

In the Make.com router the same threshold is applied as numeric filters on amount: approval route amount ≥ 100000; auto-recorded route amount < 100000.

A.3 Scenario 3 Schedule

Every Monday at 08:00 — Africa/Lagos (switched on for production use; executed on demand during testing)

A.4 Environment Variables

Variable	Use
VITE_SUPABASE_URL	Lovable
VITE_SUPABASE_ANON_KEY	Lovable
SUPABASE_SERVICE_ROLE_KEY	Make.com HTTP modules only
MAKE_SCENARIO1_WEBHOOK	Supabase INSERT webhook
MAKE_SCENARIO2_WEBHOOK	Supabase UPDATE webhook
OpenRouter API Key	Make.com OpenRouter module

A.5 Placeholder Convention

Use REPLACE_WITH_... for environment-specific values such as REPLACE_WITH_MANAGER_EMAIL and REPLACE_WITH_SUPABASE_PROJECT_REF.

A.6 Live Application

Login page: https://expense-ai-pro-14.lovable.app/login

Test username	Test password
manager@test.com	test@1234

These credentials are for evaluating the application only.

Appendix B — Final Evidence Checklist

☑ Cover page and document control

☑ System architecture diagram (Figure 1)

☑ Supabase expenses table (Figure 2)

☑ User profiles (Figure 3)

☑ Receipt storage (Figure 4)

☑ Database webhooks (Figure 5)

☑ Login page (Figure 6)

☑ Submit Expense form (Figure 7)

☑ Missing-receipt warning (Figure 8)

☑ Employee My Expenses (Figure 9)

☑ Manager Overview dashboard (Figure 10)

☑ Pending Approvals (Figure 11)

☑ Scenario 1 full canvas (Figure 12)

☑ Supabase PATCH (Figure 13)

☑ Approval route test (Figure 14)

☑ Auto-recorded route test (Figure 15)

☑ Manager approval email (Figure 16)

☑ Employee confirmation email (Figure 17)

☑ Scenario 2 full canvas (Figure 18)

☑ Duplicate-notification guard (Figure 19)

☑ Approved outcome (Figure 20)

☑ Rejected outcome (Figure 21)

☑ Needs Clarification outcome (Figure 22)

☑ Airtable mirror (Figure 23)

☑ Scenario 3 full canvas (Figure 24)

☑ By Category sheet (Figure 25)

☑ By Department sheet (Figure 26)

☑ Weekly Summary sheet (Figure 27)

☑ Weekly management email (Figure 28)

☑ Make.com scenarios dashboard (Figure 29)